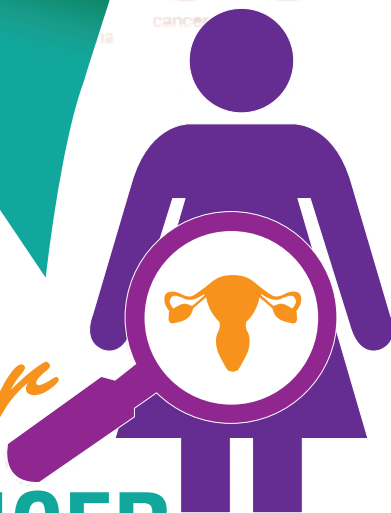




MINISTRY OF HEALTH MALAYSIA



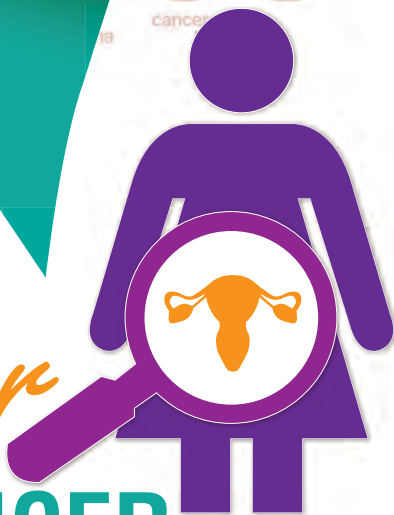
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CERVICAL CANCER SCREENING

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GUIDEBOOK *for*



CERVICAL CANCER SCREENING



GUIDEBOOK *for*

**CERVICAL CANCER
SCREENING**

2020 REFLECTION

FOREWORD

DR HAJAH FARIDAH BINTI ABU BAKAR

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In Malaysia, cervical cancer is the third most common cancer among women and the seventh highest among the entire population (Malaysia National Cancer Registry report (MNCRR), 2007 – 2011). There were a total of 4,352 cases diagnosed in 2007 to 2011 with an incidence rate (ASR) of 7.6 per 100,000 population accounting for 3.8% of all female cancers. More than 76% were diagnosed at stage II and above, where intensive and invasive treatment are required more often, culminating in increased economic burden.

Cervical cancer screening in Malaysia via conventional Pap smear was initiated in 1969 through a family service package. Decades later with relentless efforts and political advocacy, a nationwide, opportunistic, clinic-based screening, “The National Pap Smear Screening Programme” was established in 1998, aimed for early detection of cervical cancer and ensuring early treatment as well as follow-up after positive test results. This programme is still ongoing with further enhancement to ensure the processes involved are performed efficiently.

The Ministry of Health (MOH) has introduced an alternative method to address the shortcomings of the conventional Pap smear which is liquid-based cytology (LBC) in several states in the country. It has been proven that this method has higher sensitivity rate in detecting precancerous lesions. Research have shown significantly reduced numbers of unsatisfactory smears using LBC compared with the conventional Pap smear.



The contributors to this guide book has established algorithms for managing women with abnormal Pap smear and HPV results as well as management of biopsy confirmed pre-invasive lesions that can be used by all levels of healthcare workers involved in the cervical cancer screening programme. It is produced based on local experiences and references from practices of other countries such as United Kingdom, United State, Singapore and Australia. The aims of this guideline are to standardize the cervical screening programme, the management of abnormal cervical smear and biopsy confirmed cervical pre-invasive lesions.

The development of this comprehensive guideline as well as the transformation of the current cervical cancer screening tool from Pap smear to HPV test is the way forward towards enhancing early detecting of cervical cancer. It is hoped that this revised guideline will improve the programme's performance and assist in reducing the incidence of cancer of the cervix, and ultimately ensure the health and well-being of women in Malaysia.

I would like to congratulate and thank the editors and the contributors for their efforts towards the development of this revised guideline as well as those preceding it.



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LIST OF ABBREVIATIONS

AC	Adenocarcinoma
AGC-N	Atypical Glandular Cells-Favour Neoplasia
AGC-NOS	Atypical Glandular Cells-Not Otherwise Specified
AIS	Adenocarcinoma In Situ
ALTS	Triage Study For Cervical Cancer
ASC-H	Atypical Squamous Cells-Cannot Exclude High-Grade Squamous Intraepithelial Lesion
ASCUS	Atypical Squamous Cells Of Undetermined Significance
CIN	Cervical Intraepithelial Neoplasia
ECC	Endocervical Curettage
HPV	Human Papillomavirus
HR-HPV	High-Risk HPV
HSIL	High-Grade Squamous Intraepithelial Lesion
LEEP/ LLETZ	Loop Electrosurgical Excision Procedure/Large Loop Excision Of The Transformation Zone
LSIL	Low-Grade Squamous Intraepithelial Lesion
Pap smear	Papanicolaou Smear
SCC	Squamous Cell Carcinoma
TZ	Transformation Zone

1. INTRODUCTION

Cervical cancer is the fourth most common cancer worldwide, affecting women after breast, colorectal, and lung cancers, and ranks the eighth overall, with an estimated 570,000 new cases in 2018 (Bray et al., 2018). It is also the fourth most common cause of cancer death with an estimation of 311,365 deaths in 2018 worldwide (World Health Organization, 2018).

In Malaysia, cervical cancer is the third most common cancer among women and the seventh highest among the entire population (Azizah, Nor Saleha, Noor Hashimah, Asmah, & Mastulu, 2016). Chinese women were found to have the highest incidence of cervical cancer followed by Indian and Malay. Unfortunately, majority of patients (76%) with cervical cancer are often detected at stage 2 and above which affects their prognosis.

A total of 4,352 cervical cases were diagnosed between 2007 – 2011 (ASR was 7.6 per 100,000 population) which accounted for 3.8% of all female cancers (Azizah et al., 2016). Cervical cancer incidence rate increased with age after 30 years and has its peak at ages 60–69 years (Malaysian Health Technology Assessment Section (MaHTAS), 2011).

Therefore, it is imperative to ensure the implementation of screening in assuring early detection of cervical cancer. The cornerstone of cervical cancer screening worldwide has been the Papanicolaou test; also known as the Pap test or the Pap smear, which was developed by Georgios Papanikolaou in 1940s. This tool is deemed to be useful in identifying morphological changes in the cervical epithelium. The slow evolvement of this disease through a long latent period supports its early detection. Thus, diagnosing cervical cancer at an early stage and providing access to effective treatment can significantly improve the likelihood of survival.

Screening of cervical cancer was introduced in 1969 in Malaysia, as part of a family services package, through the Maternal and Child Health Services, Ministry of Health Malaysia. Conventional Pap smear was the screening method employed in the clinics. Cervical cancer gained further

leverage in 1995 when 'cancer' was chosen to be the theme of the Healthy Lifestyle Campaign.

The objectives of this cervical cancer screening programme are:

- a. To ensure early detection of cervical cancer
- b. To determine the magnitude and extent of cervical cancer among the target group
- c. To plan activities for management and follow-up of women with abnormal Pap smears

The current national policy for Pap smear screening programme is to screen all women aged 30 and 65 years for cervical cancer, once every three years.

The programme is carried out as an opportunistic screening where women who attend the clinics for health screenings will be offered to undergo Pap smear screening. Other agencies that are providing this service include the National Population and Family Development Board (NPFDB), university hospitals, private hospitals and clinics, hospitals and health facilities of Ministry of Defence and the non-governmental organizations (NGO) for example Federation of Reproductive Health Association Malaysia (FRHAM), National Cancer Society of Malaysia (NCSM) and National Cancer Council (MAKNA).

Pap smear coverage in Malaysia was 26% in 1996 (Narimah, 1997) and 43.7% in 2006 (Institute for Public Health (IPH), 2008). Most of these Pap smears are done on women who come for post-natal check-up (Nor Hayati & Ayob, 1997). Despite this, primary cervical cancer screening by Pap smear is considered as the most successful cancer screening programme to date.

Another technique of Pap test is Liquid-based Cytology (LBC) which offers an alternative of cervical cell sample preparation to address the shortcomings of the traditional Pap test. It produces better quality slides since unwanted materials such as blood clots or mucous are removed as well as cells are more evenly distributed. Studies has demonstrated that

this technique is as effective as the traditional smear (Moseley & Paget, 2002; Strander, Andersson-Ellström, Milsom, Rådborg, & Ryd, 2007).

Despite its success, cytology has limitations especially technical limitations regarding sampling and laboratory errors in screening and interpretation. Therefore in recent years, there has been interest for producing new tests with acceptable sensitivity and specificity for detecting clinically significant cervical cancer precursors. One method is HPV testing based on the knowledge that it results from Human Papillomavirus infection.

This test will be incorporated in the current cervical cancer screening programme. In contrast to screening by cytology, HPV testing is based on the detection of HPV DNA, HPV mRNA or other viral markers (Chrysostomou, Stylianou, Constantinidou, & Kostrikis, 2018). This test maybe done as a primary screening strategy every 5 years or combined with Pap smear in women over 30 years of age at an interval of at least 3 to 5 years in women who are negative on both tests in the annual screening (Malaysian Health Technology Assessment Section (MaHTAS), 2011).

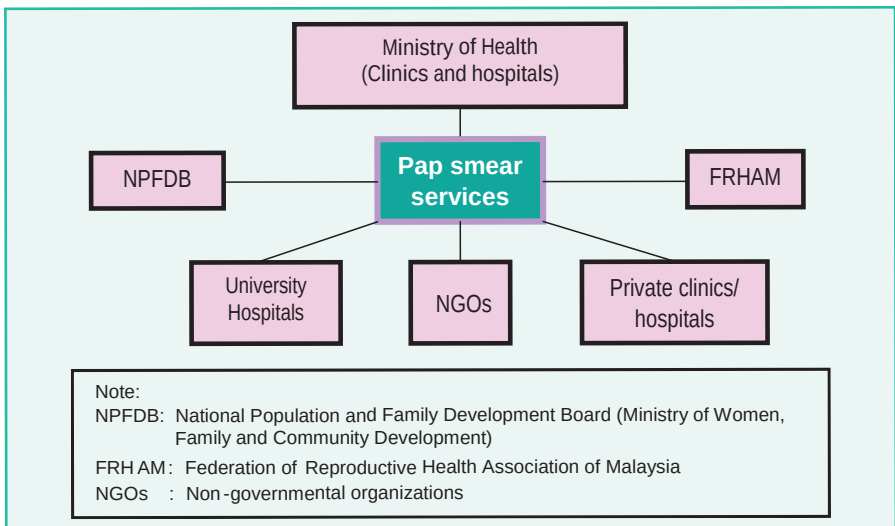


Figure 1: Organizations Providing Cervical Cancer Screening Services



2. OBJECTIVES

- 2.1. As a guide and reference for health personnel
- 2.2. To standardize the cervical screening programme
- 2.3. To standardize the management of abnormal cervical smear
- 2.4. To standardize the management of biopsy confirmed cervical pre-invasive lesion.

3. SCREENING POLICY

3.1. Target groups

The cervical smear services are offered to all sexually active women between the ages of 30 and 65 years. However, those who are sexually active but less than 30 years are encouraged to come for screening.

3.2 Screening interval

- The initial screening interval is yearly for two years. If the results were normal, then a 3-yearly Pap smear screening is indicated.
- If cytology and HPV testing were done together and both results were negative, then the cervical screening interval is 5-yearly.

3.3 Screening Personnel

Doctors and nurses trained in taking cervical smears.

3.4 Reporting Personnel

Pathologists and Cytotechnologists.

4. SCREENING OPTIONS

4.1. Cytology Based Screening

4.1.1 Conventional Pap smear

Sample from cervical scrape is obtained using cervical brush/broom and smeared directly onto a glass slide, immediately fixed with 95% alcohol and stained with Papanicolaou method for microscopic examination.

4.1.2 Liquid-based Cytology

Sample from cervical scrape is obtained using cervical brush/broom and suspended in a vial of preservative for transport to the laboratory.

The sample is processed and placed onto a glass slide and stained with Papanicolaou method for microscopic examination. The choice of technology must be FDA approved or equivalent.

4.2 Molecular based technique using HPV test

HPV test is a molecular technique using a sample of cervical cells, to detect the presence of Human Papillomavirus (HPV). This test screens for high risk HPV genotypes. HPV test has been validated as reflex testing for ASC-US and Low- Grade Squamous Intraepithelial Lesion.

Many developed countries have adopted HPV testing as a primary screening tool (Pan American Health Organization (PAHO), 2016). Incorporation of HPV testing into cervical cancer screening programmes has the possibility of increased disease detection and lengthening the screening interval (Saslow et al., 2012).

4.3. Others

In low resource areas, other options of cervical screening can be adopted such as Visual Inspection with acetic acid (VIA)/VILLI.



5. TAKING A PAP SMEAR

5.1. Advice for the woman

- i. Do not douche or insert any form of medication (suppositories) into the vagina 24 hours prior to the procedure.
- ii. Avoid sexual intercourse within a period of 24 hours prior to the procedure.
- iii. Avoid taking Pap smear during menstruation.
- iv. Pap smear can be done during pregnancy.

5.2. Requirements for Taking Smears in Clinic

- i. Bivalve vaginal speculum (CUSCO) - all sizes preferably disposable.
- ii. Swabs
- iii. Normal saline
- iv. Hand gloves
- v. Sampling devices or instruments (refer to **appendix C**)
- vi. Request form - PS1/98 (Pindaan 2019)
- vii. Adequate light source
- viii. Couch and screen

5.2.1 Conventional Smear

- i. Frosted end glass slide and 2B pencil or ordinary glass slide and diamond pencil.
- ii. Fixative – 95% ethyl alcohol in coplin jar or alcohol spray
- iii. Slide mailer

5.2.2 Liquid-based Cytology

Preservative vial

5.3. How to take a Pap smear

For further clarification, please refer to **Figure 2**.

1. Complete the cytology request form - PS 1/98 (Pindaan 2019) and label the glass slide or vial (**Figure 3**).
2. Wash your hands and wear gloves.
3. Examine the woman in a dorsal position.
4. Swab the introitus with normal saline.
5. Wet the bivalve speculum using sterile water or normal saline (Do not use lubricant).
6. Introduce the speculum into the vagina carefully avoiding contact with the cervix (for conventional smear, bleeding from the cervix will interfere with the evaluation of smear).
7. Expose the cervix clearly. If there is a discharge, take a swab for microscopic examination and send for culture and sensitivity if indicated. Remove excess discharge before taking the cervical sample.
8. Take a sample from the cervix using the sampler device according to the sample type:
 - i. Conventional:
Take sample from the cervix including transformation zone, turn one complete circle 360 degrees. Smear the sample onto the labelled glass slide in one direction (**Figure 5**).

Note:

- i. Position of transformation zone (TZ) varies according to age. Selection of sampling device should be in accordance to the location of transformation zone. (Figure 8A and Figure 8B)
- ii. Smear should be taken before performing bimanual examination
- iii. During colposcopic examination, smear should be taken before applying acetic acid

- ii. Liquid-based:
Take sample from the cervix including transformation zone.
Follow the instruction as in user manual, example:

ThinPrep: 360 degrees rotation 3X.

SurePath: 360 degrees rotation 5X in clockwise manner.

9. Fixation:

- i. Conventional
 - a) After taking smear, immediately dip the slide into coplin jar containing 95% ethyl alcohol for 30 minutes. Once completed, stand the slide on slide rack to drain excess fluid.
 - b) For spray fixation, spray fixative should be placed 15-25cm and sprayed at the right angle. (Refer **Figure 6**)
- ii. Liquid-based
Follow instruction as in user manual, example
 - a) ThinPrep:
Rinse the brush vigorously inside the preservative vial by swirling and pushing against the vial wall 10 times.
Discard the brush.

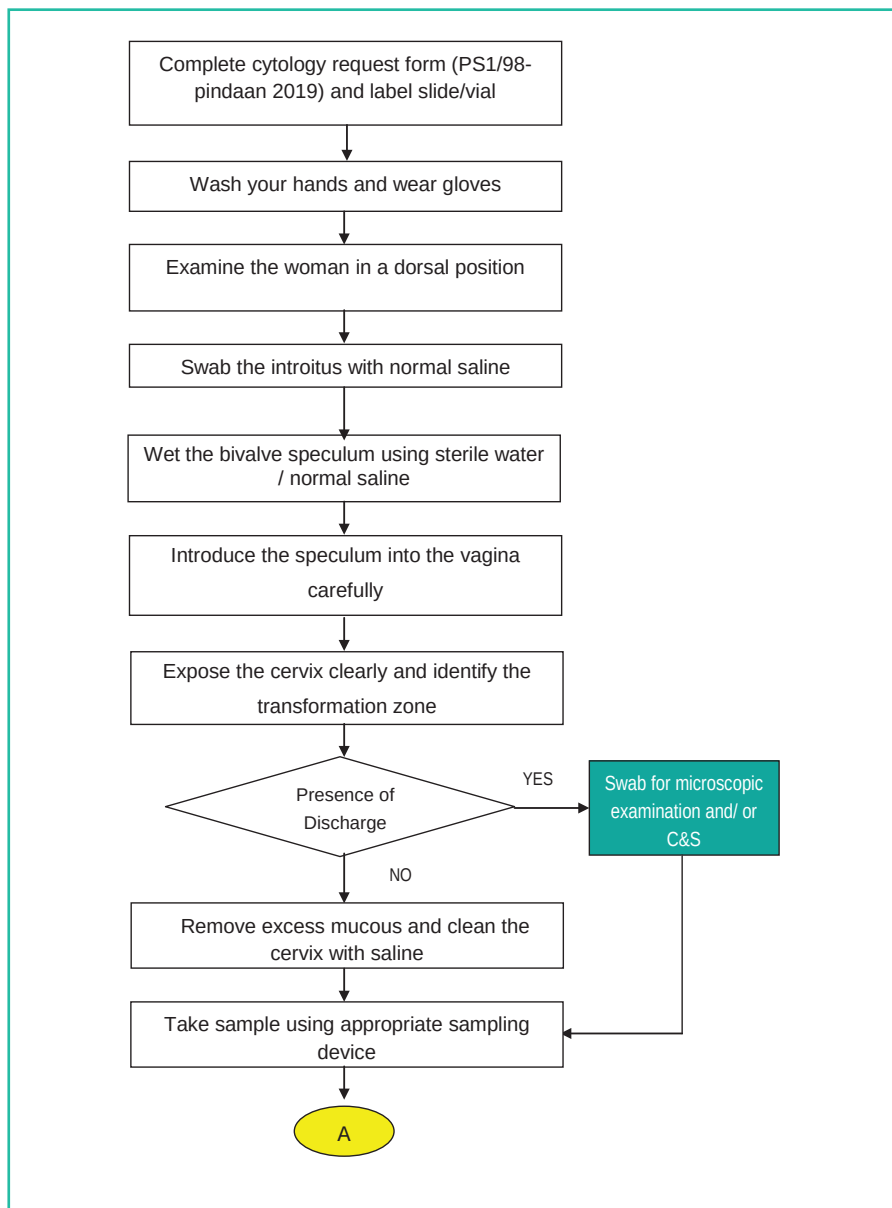
Note:

- The fixative (95% ethyl alcohol) must be placed in a covered container to avoid evaporation.
- The fixative must be changed regularly
- Do not allow the smeared slide to dry before fixation
- Endobrush may be used in severely atrophic cervix
- Ensure correct labelling of slide and completion of form
- Details of previous smears (if relevant) must be stated in the request form

- #### 5.4 Reasons for unsatisfactory smear

- Insufficient sample
- Inadequate fixation time (less than 30 minutes)
- Delay in dipping the slide in the fixative
- Blood-stained smear
- Thick smear
- Excessive discharge (on the slide)
- Broken slide
- Usage of lubricant before taking smear
- Dirty and contaminated slides

HOW TO TAKE A PAP SMEAR:



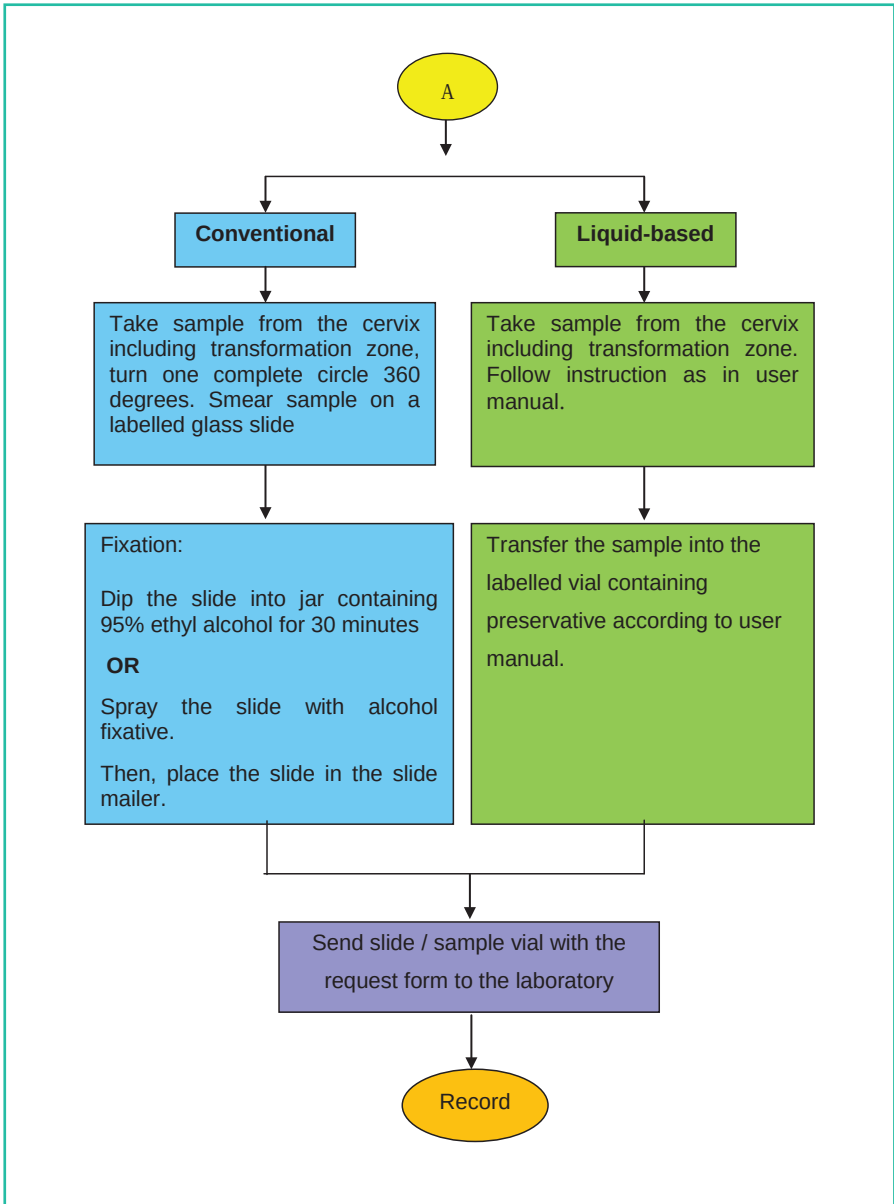


Figure 2: Pap Smear Sampling Procedure

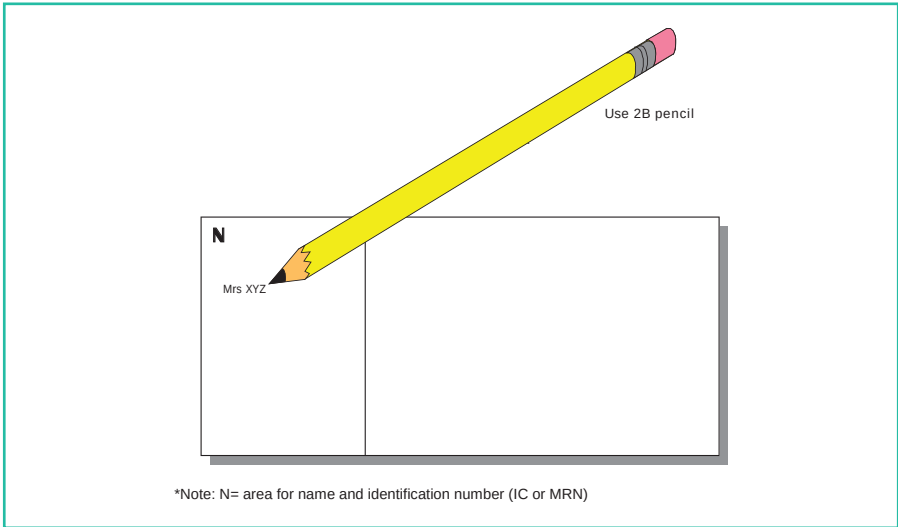


Figure 3: Labelling with 2 unique identifiers on the frosted end of glass slide

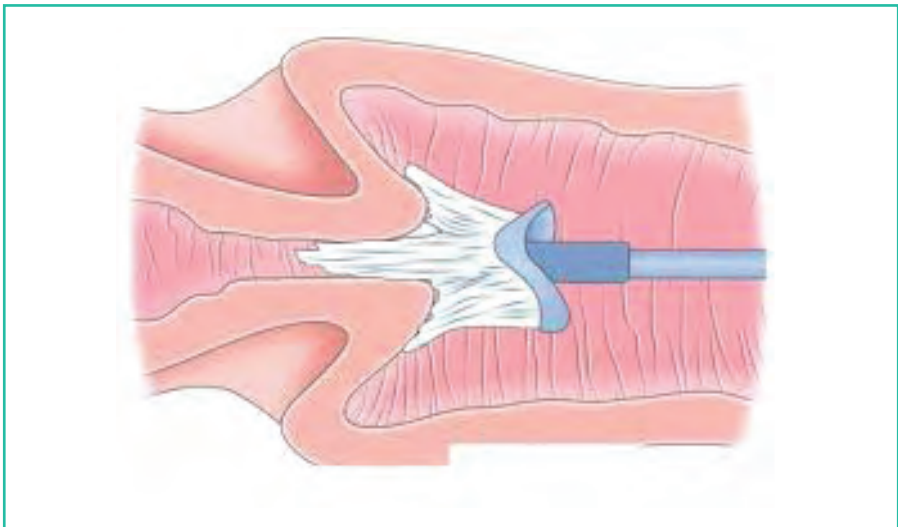


Figure 4: Cervical Smear Using Cervical Sampler Broom



Figure 5: Smearing the sample using a cervical sample broom in one direction

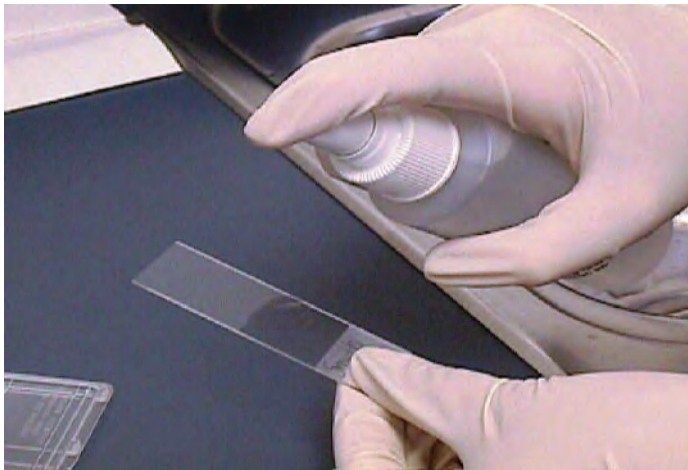


Figure 6: Fixation of smear by alcohol spray

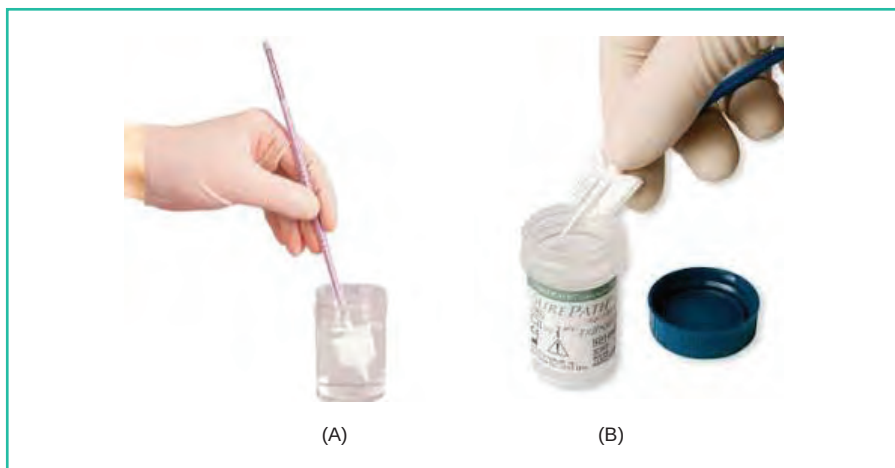


Figure 7: Transferring of sample into the vial containing preservative

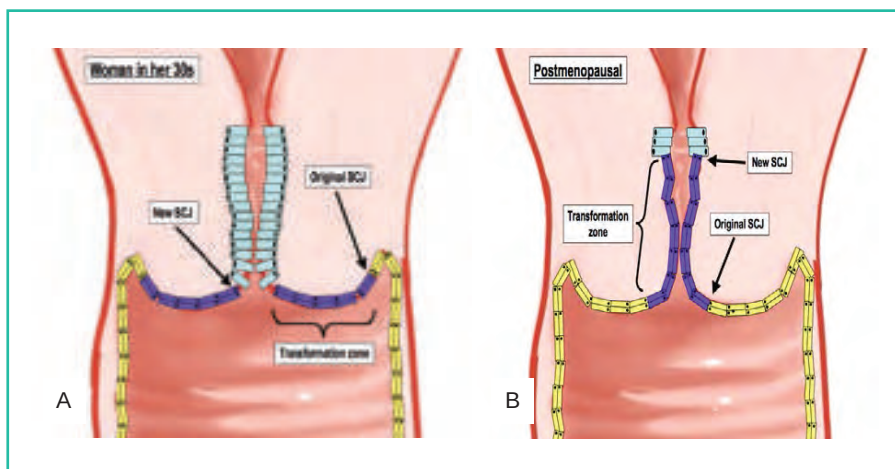


Figure 8: Position of transformation zone (TZ). Figure 8 (A) is TZ for reproductive age group in which cervical sampler broom is recommended. Figure 8 (B) is TZ for post-menopausal women in which both cervical sampler broom and endocervical brush can be used.

6. SAMPLE PROCESSING IN THE CYTOLOGY LABORATORY

Cytology laboratory shall be equipped with adequate resources for example equipment, dedicated, trained cytotechnologists and anatomical pathologists. The laboratory shall participate in quality assurance program and comply with the International Standard of Medical Laboratory Accreditation (ISO 15189).

6.1. Equipment

- i. Slide staining rack
- ii. Staining trough
- iii. Pap stains manual or automated
- iv. Sink with tap
- v. Slide drying stand rack
- vi. Mounting medium
- vii. Cover slip
- viii. Binocular light microscope –
lenses with objectives X10, X20, X40, X60 (optional)
- ix. Slide labels
- x. Slide filing cabinet
- xi. Fume hood for mounting
- xii. Liquid-based slide processor/s

6.2. Processing

6.2.1 Conventional

Glass slide will be stained with Papanicolaou method (manual/ automated) and examined microscopically for any abnormality. Screening is done manually by a cytoscreener with stringent quality assurance programme in place to maintain quality result which includes 100% rescreeing (by 2 another cytoscreener) and targeted screening for abnormal clinical history.

6.2.2 Liquid-based

Sample vial will be processed using processor/s. This method produces an evenly spread cells with cleaner background. Screening is done manually by a cytoscreener as for conventional smear. Automated screening system is an available option as a primary screening method. However, it is required to be validated manually by cytoscreener/pathologist.

6.2.3 HPV testing

Various HPV DNA testing methods are available. Most are suitable to be used with liquid-based sample. To maintain the validity of this sample, the remaining sample vial should be kept in an optimal condition.

7. PAP SMEAR REPORTING

Pap smears are reported using The Bethesda Reporting System 2014.

7.1 Specimen Type

Indicate conventional smear or liquid-based preparation

7.2 Adequacy of specimen

7.2.1 Satisfactory for evaluation

It is pertinent to describe presence or absence of endocervical/ transformation zone component and any other quality indicators e.g. obscuring blood, inflammation etc.)

7.2.2 Unsatisfactory for evaluation

The reasons for unsatisfactory evaluation must be specified for example scanty squamous cells.

- 
- Atypical
 - Endocervical cells, favor neoplastic
 - Glandular cells, favor neoplastic
 - Endocervical adenocarcinoma in situ (AIS)
 - Adenocarcinoma
 - Endocervical
 - Endometrial
 - Extrauterine
 - Not otherwise specified (NOS)

7.3.2.3 Other malignant Neoplasms: specify.

8. MANAGEMENT OF ABNORMAL CERVICAL SMEARS

8.1 Management of Unsatisfactory Smear

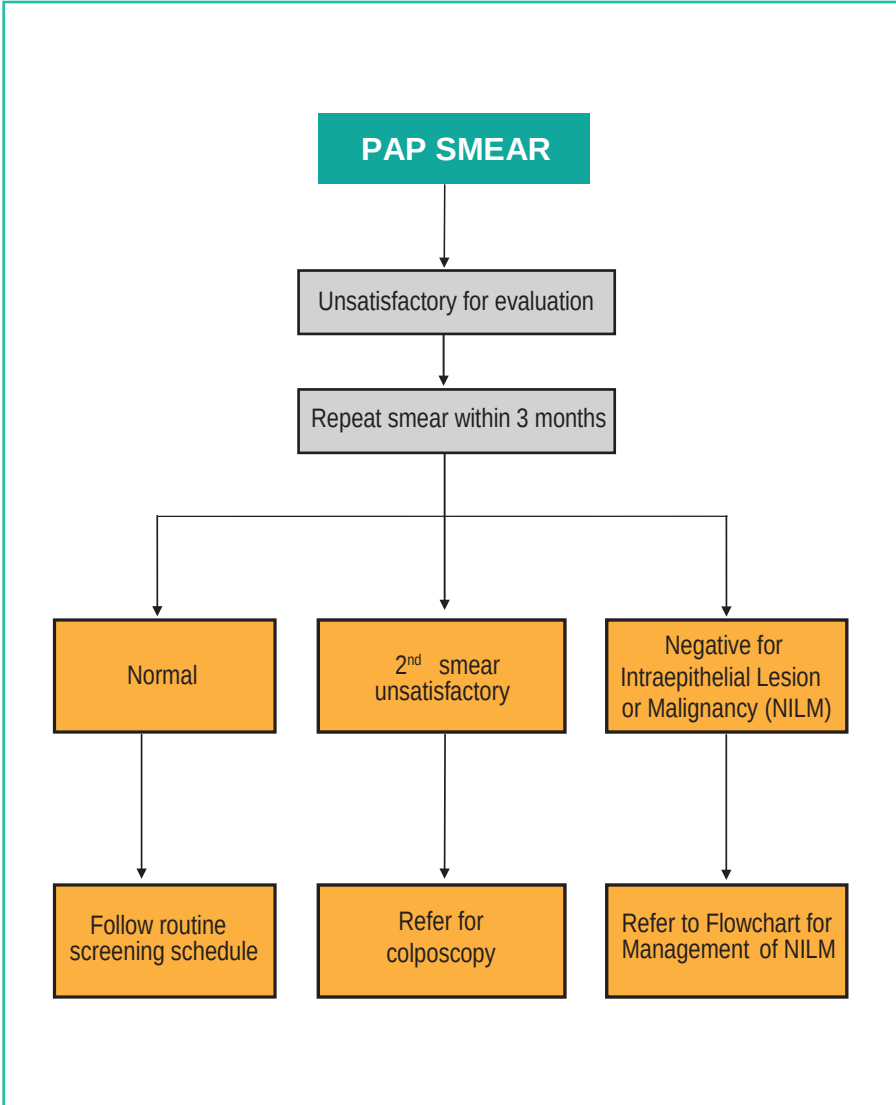


Figure 9: Management of Unsatisfactory Smear

8.2 Management of Normal Smear

8.2.1. Management of Negative for Intraepithelial Lesion or Malignancy (NILM)

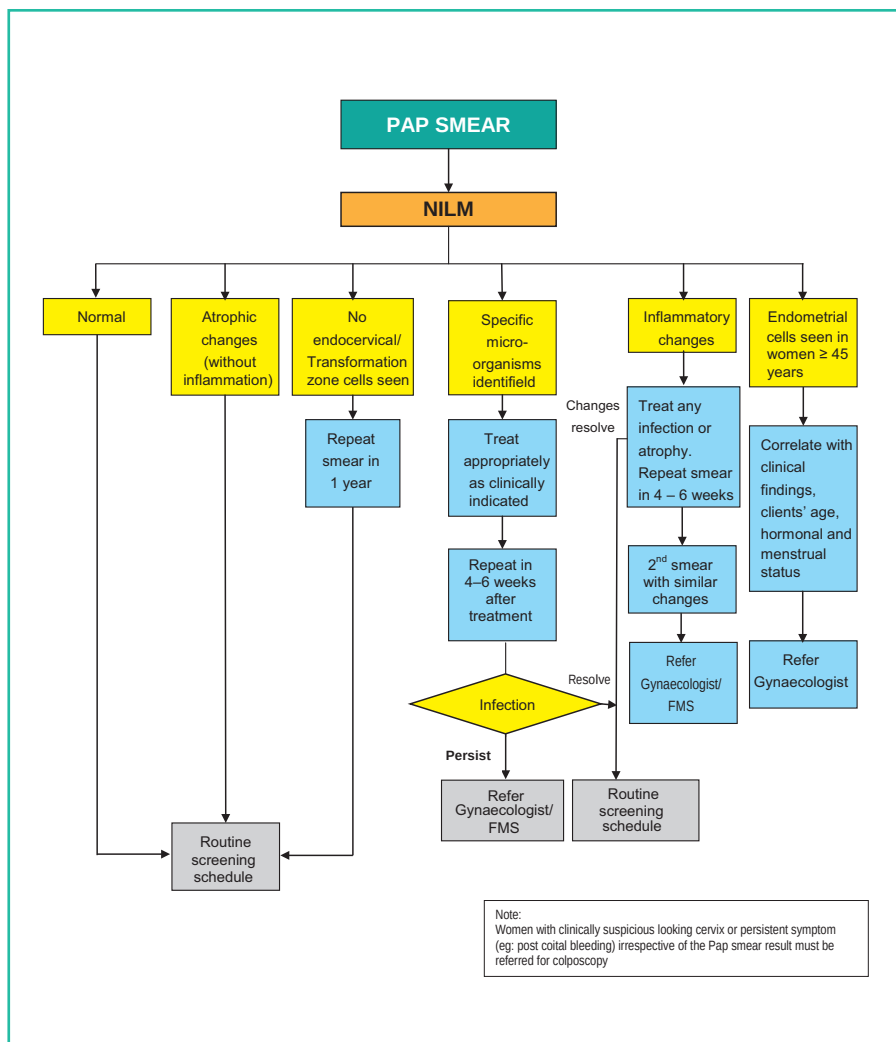


Figure 10: Management of NILM

8.3 Management of Abnormal Pap Smear:

8.3.1. Squamous Cell Abnormalities

8.3.1.1. Atypical Squamous Cells

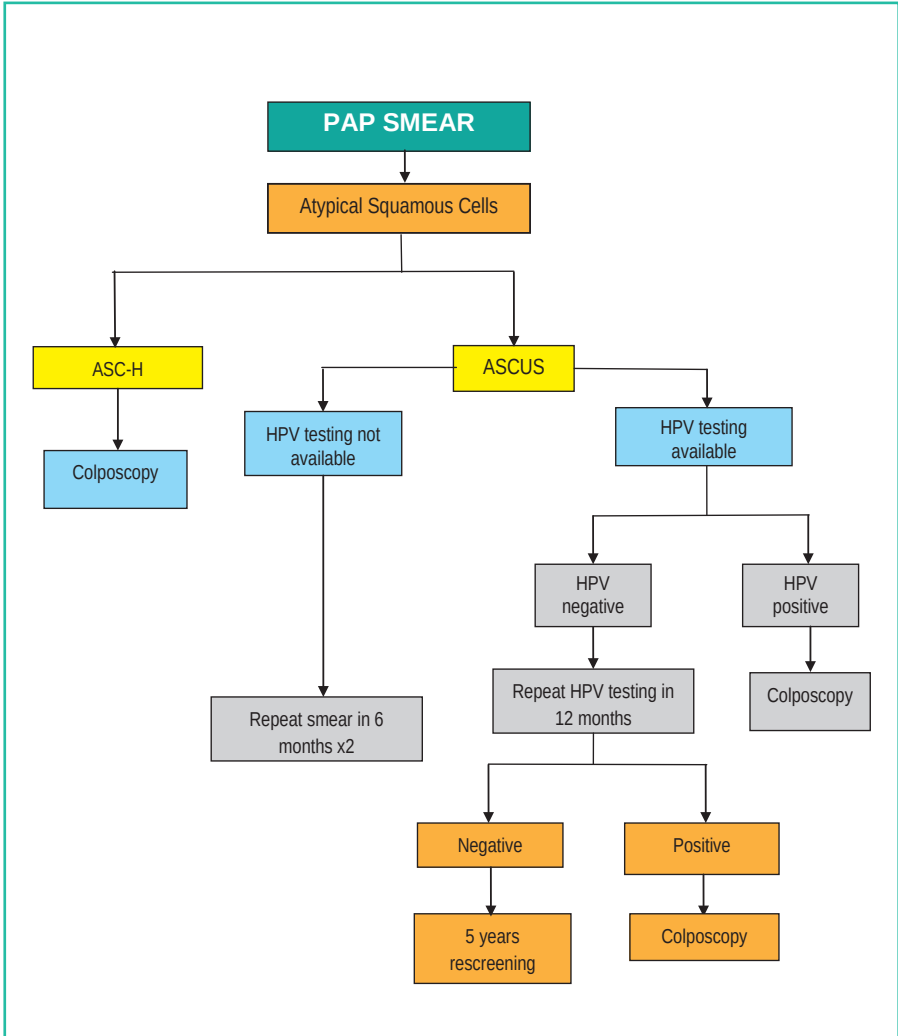
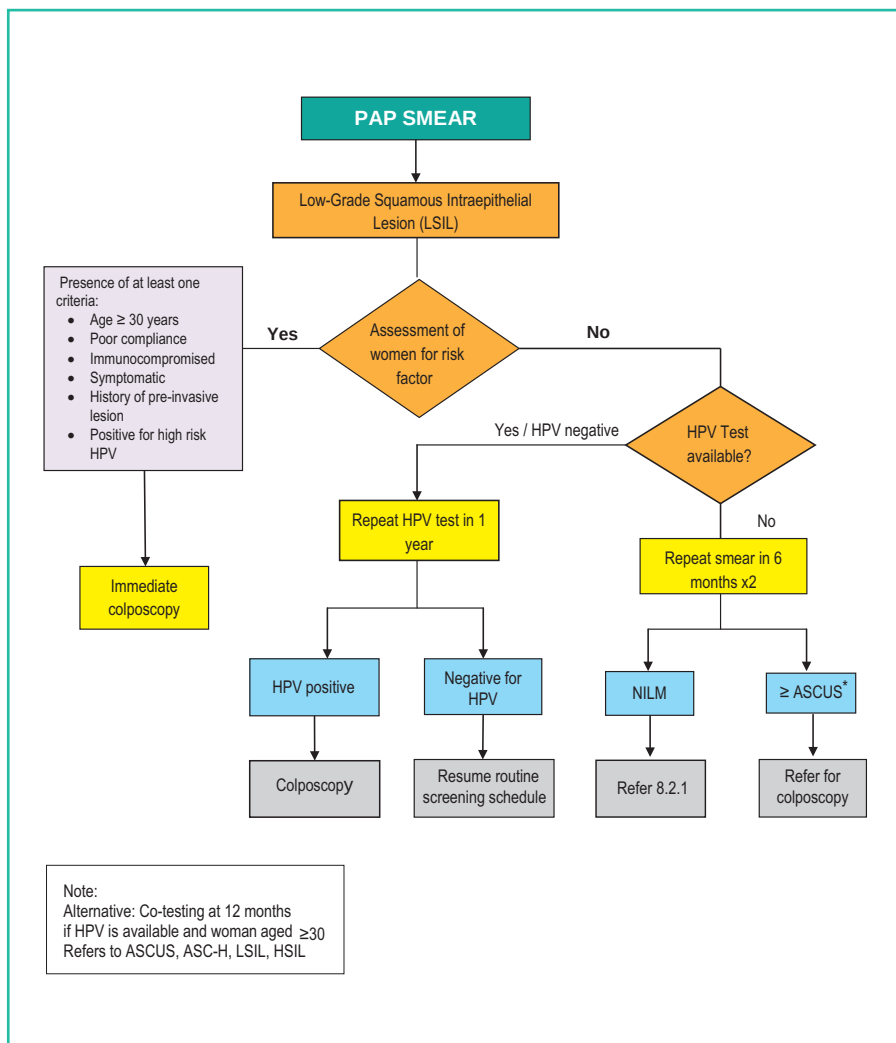


Figure 11: Management of Atypical Squamous Cells

8.3.1.2. Low-grade Squamous Intraepithelial Lesion (LSIL)

**Figure 12:** Management of Low-grade Squamous Intraepithelial Lesion

8.3.1.3. High-grade Squamous Intraepithelial Lesion (HSIL) and Squamous Cell Carcinoma

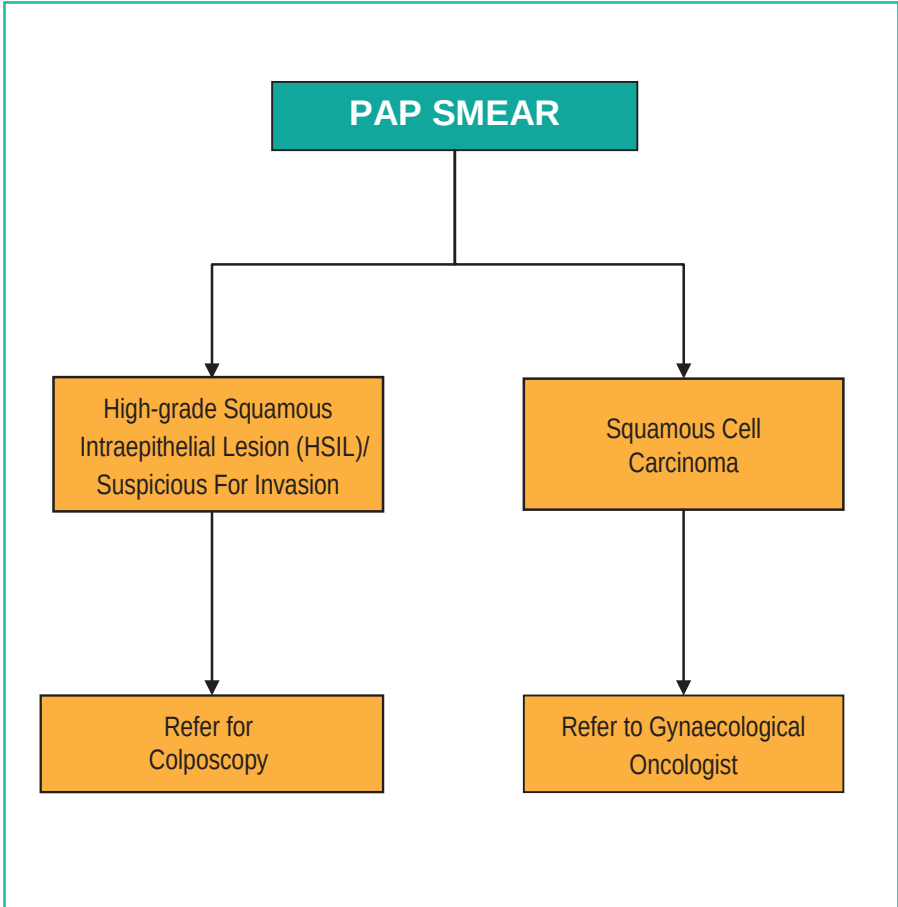


Figure 13: Management of High-grade Squamous Intraepithelial Lesion and Squamous Cell Carcinoma

8.3.2. Glandular Cell Abnormalities

8.3.2.1. Atypical Glandular Cells and Adenocarcinoma

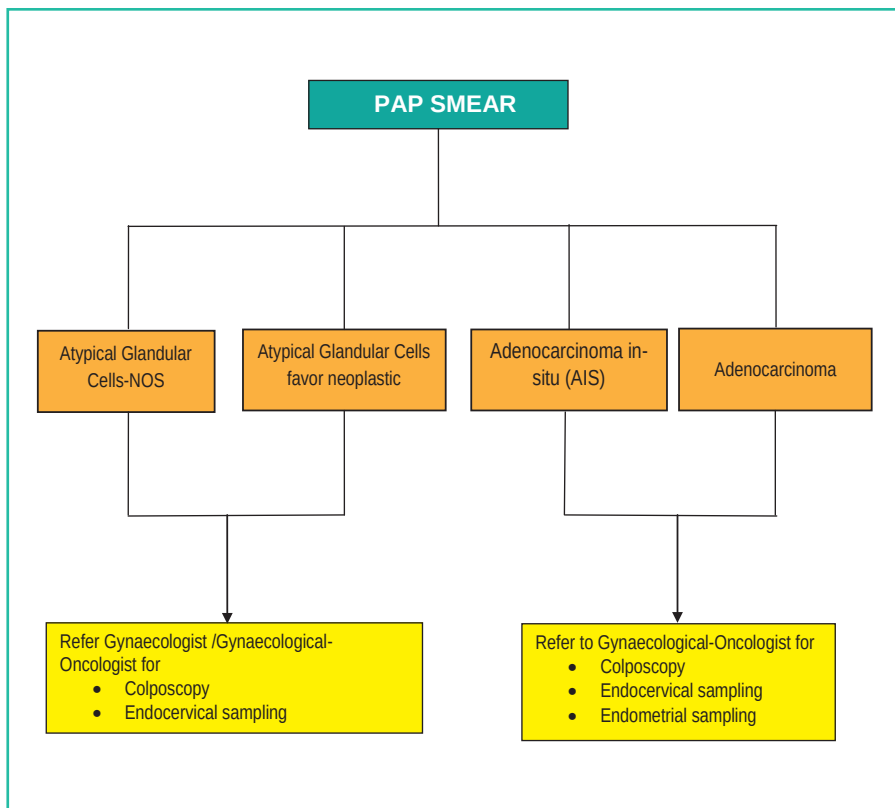


Figure 14: Management of Atypical Glandular Cells and Adenocarcinoma

8.3.2.2. Subsequent Management for Atypical Glandular Cells / Endocervical Adenocarcinoma-in-situ (AIS)/ Adenocarcinoma on Pap smear

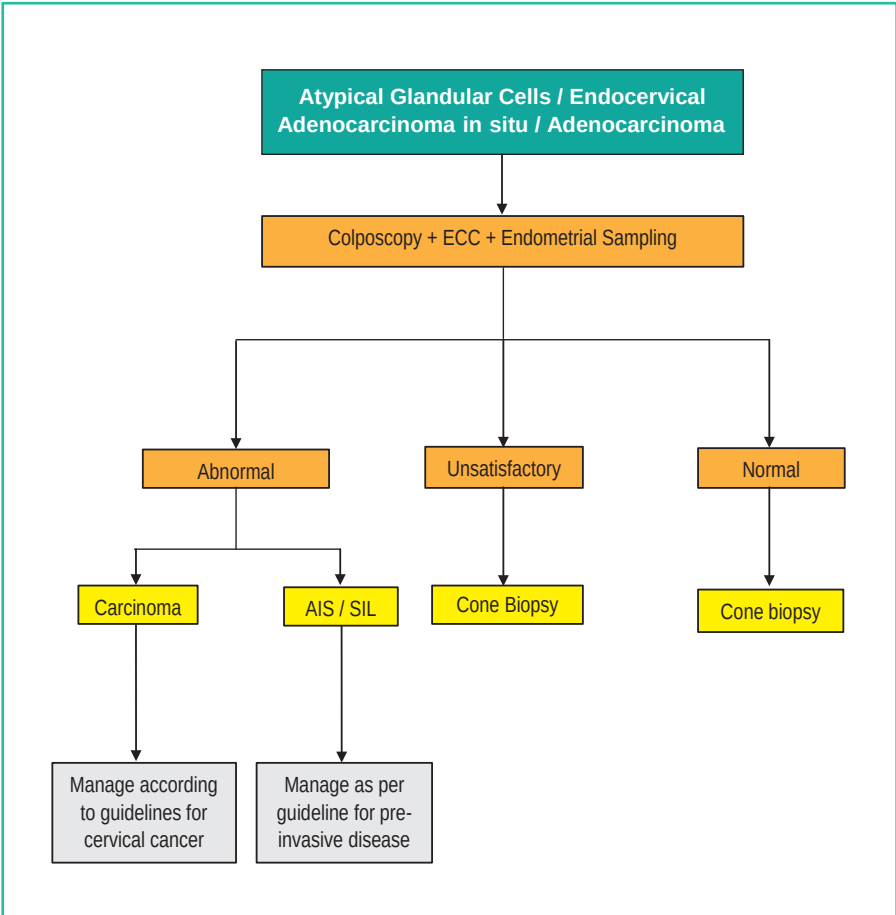


Figure 15: Management of Atypical Glandular Cells/Endocervical Adenocarcinoma in situ (AIS)/ Adenocarcinoma

8.3.3. Management of woman age ≥ 45 years with endometrial cells presence undergoing Pap smear

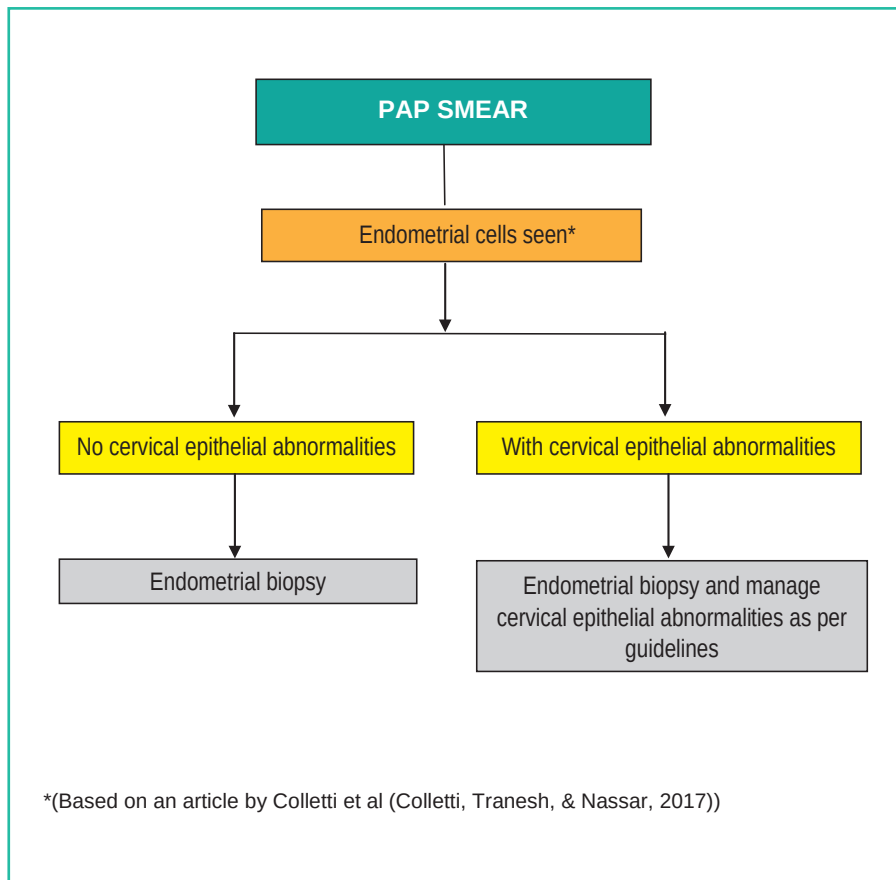


Figure 16: Management of presence of endometrial cells in women more than 45 years old

8.3.4. Management of HPV results following HPV DNA testing (lab-based test)

(Refer Guidelines for Primary HPV Testing In Cervical Cancer Screening In Malaysia)

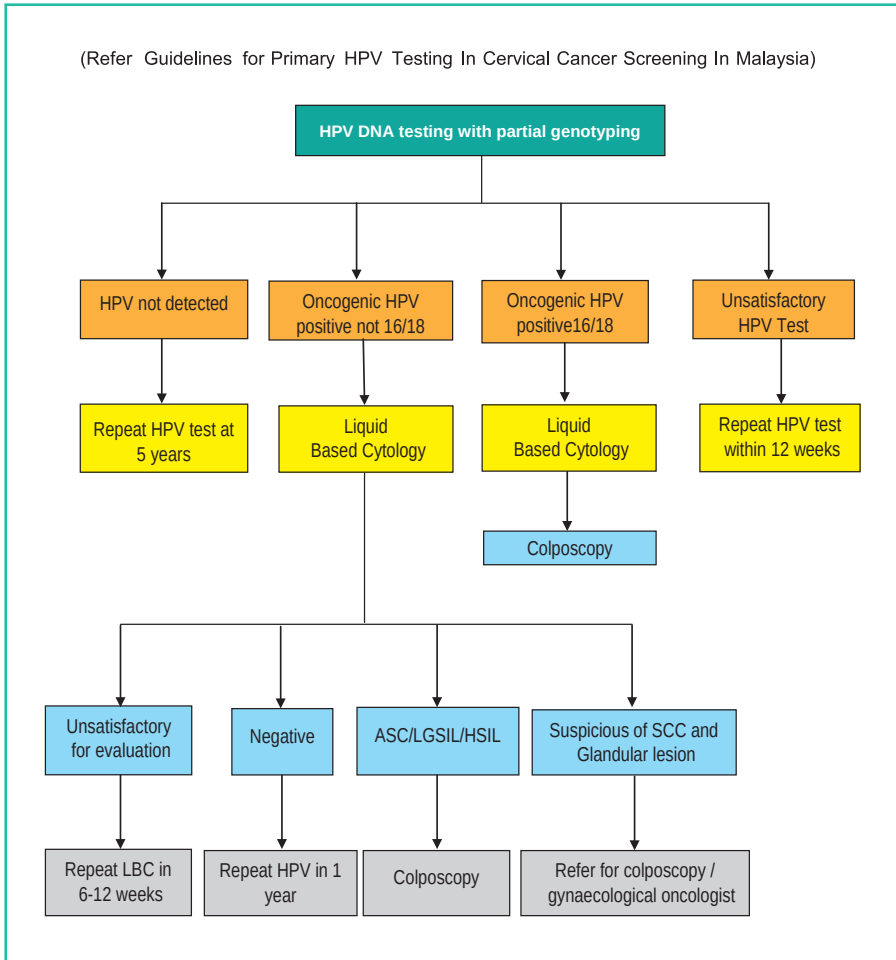


Figure 17: Management of HPV results following HPV DNA testing

8.4 Pap Smear Guidelines after Hysterectomy

Table 1: Guidelines for Pap Smear Following Hysterectomy

No.	Status	Action
8.4.1	Hysterectomy for benign disease: - Normal Pap smear history. - Histopathology of cervix known and is benign with no dysplastic / neoplastic changes	In the absence of symptoms, may not require any further screening.
8.4.2	Subtotal hysterectomy	Should continue to have Pap smears according to normal screening schedule.
8.4.3	Hysterectomy where histology is not known	Should have one baseline vault smear. If this is normal, further screening should be based on clinical indications.
8.4.4	Immunosuppressed women	Should continue to have vault smears at yearly intervals.
8.4.5	Women with past history of CIN 2 & 3: - If excision margin was involved or histological assessment inadequate	Follow-up should be at the discretion of the gynaecologist. In general, vault smears should be taken at least yearly.

No.	Status	Action
	- CIN 2 / 3 completely excised at hysterectomy regardless of margin	Vault smears yearly for 5 years followed by two yearly smears.
8.4.6	Women previously treated for invasive gynaecological malignancy	Should be followed up by a gynaecologist, preferably a Gynaecological Oncologist.

8.5 Management of Abnormal Pap Smear and CIN in Pregnancy

- Colposcopic examination should be undertaken to exclude invasive disease by a Colposcopist.
- If a high grade lesion is suspected on colposcopy, a biopsy is indicated to exclude possible invasive disease. Cervical biopsy is safe in pregnancy.
- For histology confirmed CIN 2 or 3, colposcopic review should be done in the second and third trimester to exclude any possible progression to invasive disease.
- Treatment of CIN should be deferred till at least 6 weeks postpartum, when the lesion should be reassessed.

8.6 Indications for colposcopy

The following conditions are indicated for colposcopy:

- Suspicious looking cervix
- Unexplained post-coital bleeding, blood-stained vaginal discharge and postmenopausal bleeding
- Persistent unsatisfactory smear on 2 occasions, 3 months apart
- Persistent inflammatory smear on 2 occasions, 4 to 6 weeks after treatment
- Persistent Atypical Squamous Cells of Undetermined Significance (ASC-US) on 2 occasions
- Atypical Squamous Cells of Undetermined Significance (ASC-US), positive for high risk HPV
- Atypical Squamous Cells—cannot exclude high grade lesion (ASC-H)
- Persistent Low Grade Squamous Intraepithelial Lesion (LSIL) on 2 occasions, 6 months apart
- Persistent Low Grade Squamous Intraepithelial Lesion (LSIL) with high risk factors
- High Grade Squamous Intraepithelial Lesion (HSIL)
- Squamous Cell Carcinoma (SCC)
- Atypical Glandular Cells (AGC)
- Adenocarcinoma
- Positive for high risk HPV DNA

9. MANAGEMENT OF BIOPSY CONFIRMED PRE-INVASIVE LESION

9.1 Grade 1 (CIN 1)

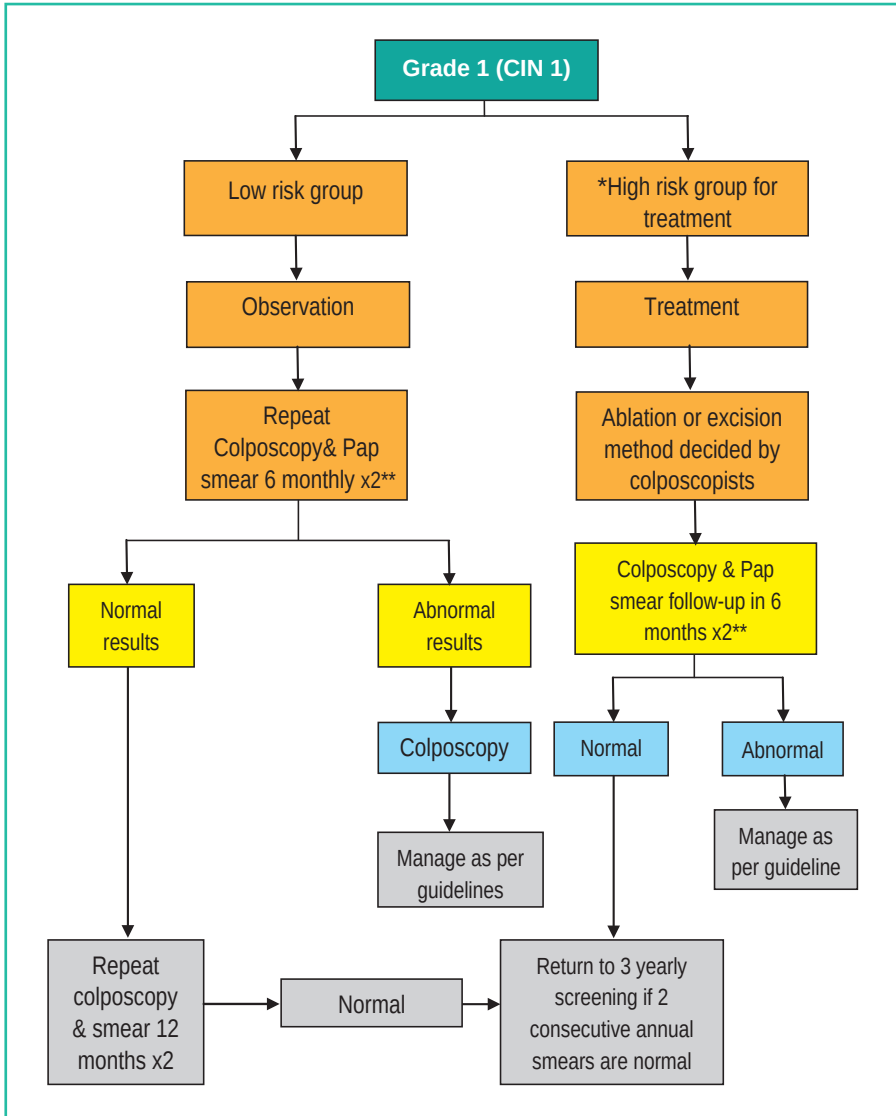


Figure 18: Management of Biopsy Confirmed Pre-invasive Lesion

* High Risk Group:

- Immunocompromised patient
- Smokers
- Multiple sexual partners
- Positive to high risk HPV
- Family history of Gynaecological Cancer
- Poor compliance for follow-up
- Age ≥ 30 years
- Poor access to health care
- Substance abuse

** Alternative:

Co-testing in 12 months if HPV testing is available and woman age ≥ 30

9.2 Grade 2/3 (CIN 2 / CIN 3)

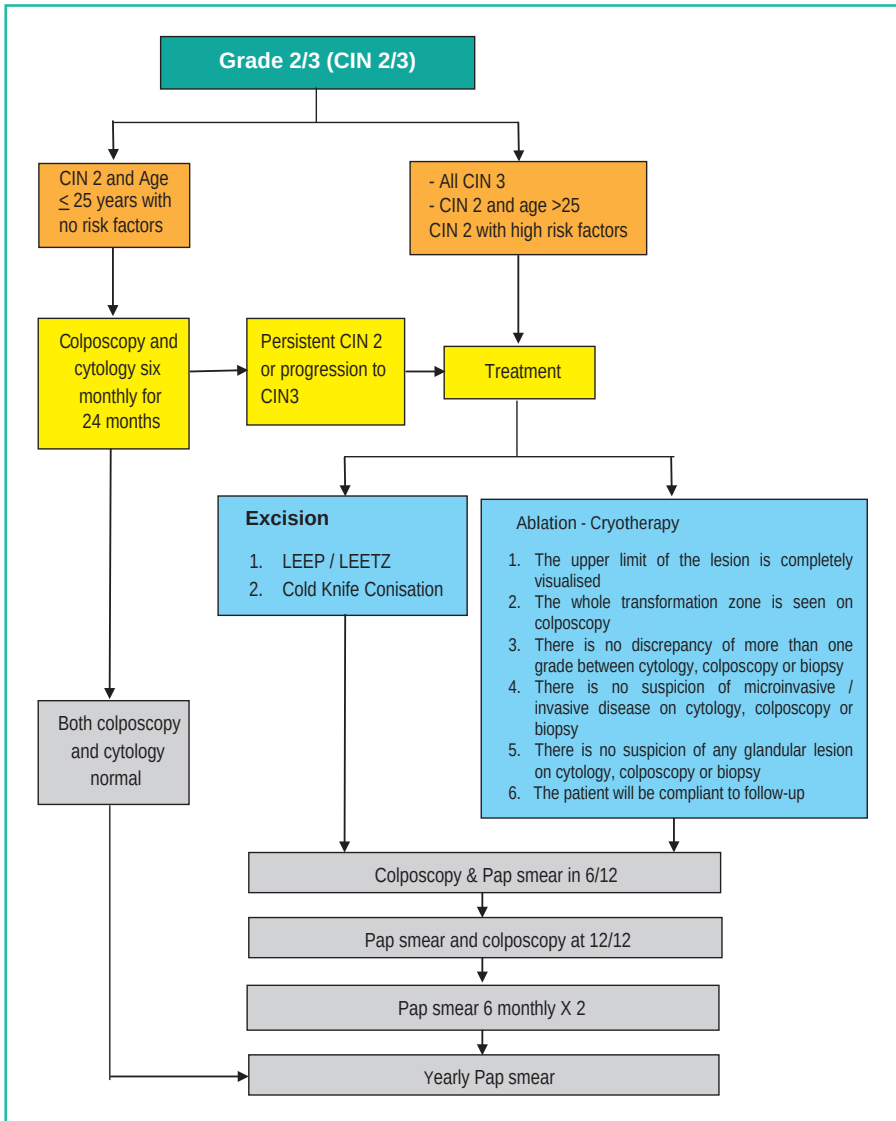


Figure 19: Management of CIN 2/CIN 3

Note

CIN 2/CIN 3:

- i. Evidence suggests that regression of CIN 2 in women less than 25 years occurs at a rate similar to CIN 1.
- ii. CIN 2 in women less than 25 years old should be observed with colposcopy at 6-month intervals for up to 24 months before treatment is considered.
- iii. CIN 3 in women less than 25 years old should be treated.

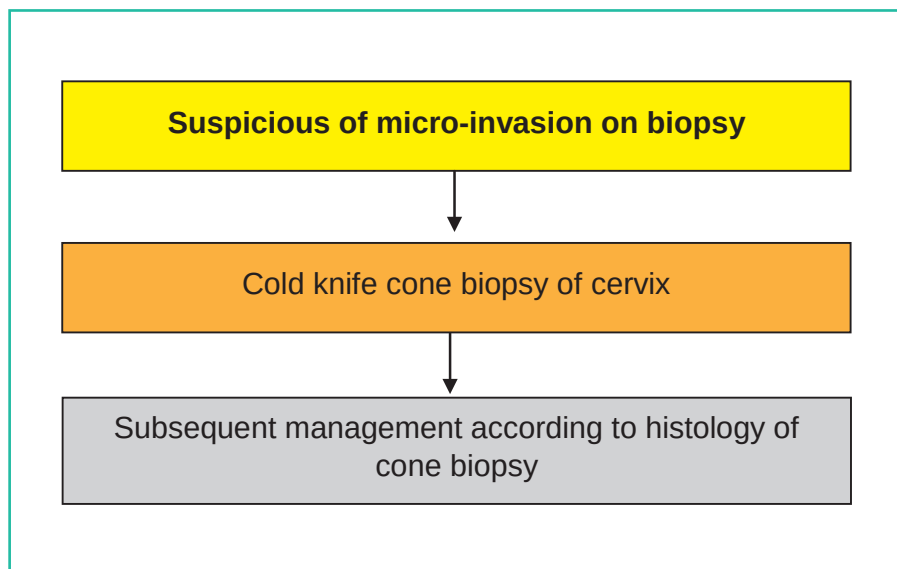
9.3 Suspicious of Micro-Invasion on Punch Biopsy

Figure 20: Management of Suspicious of Micro-Invasion on biopsy

9.4 Adenocarcinoma in-situ on punch biopsy

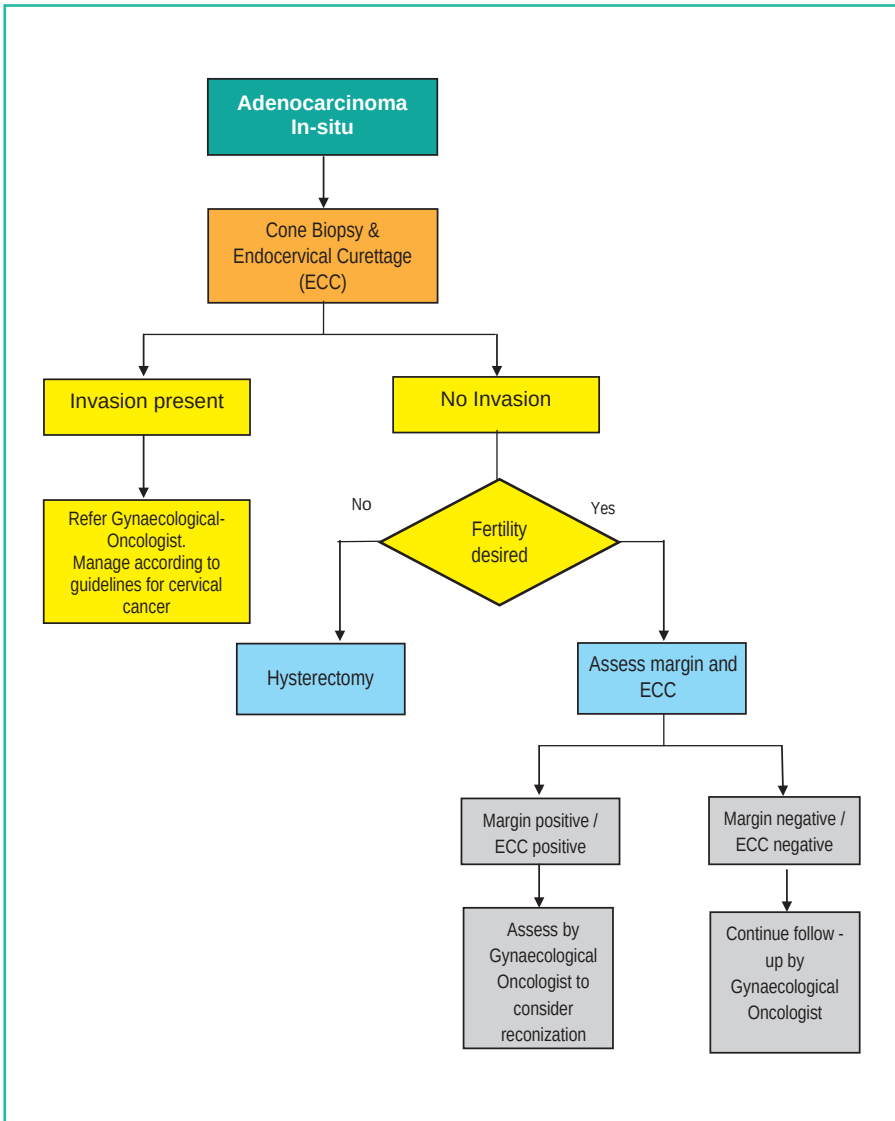


Figure 21: Management of Adenocarcinoma In-situ on punch biopsy

10. COLPOSCOPY MANAGEMENT GUIDELINES

10.1 Introduction

Terminology

Terminology is based on the 2011 IFCCP nomenclature (Bornstein et al., 2012). Satisfactory colposcopy is defined as adequate visualisation of cervix and a SCJ being fully visible, i.e. a type 1 or 2 TZ.

Type of Transformation Zone (TZ)

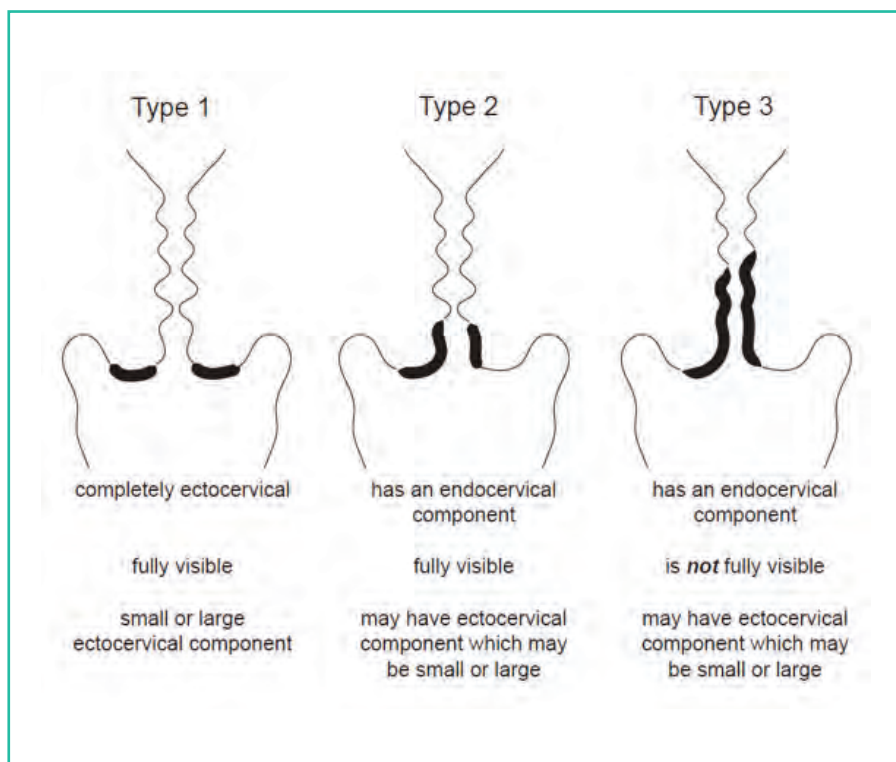


Figure 22: Transformation Zone Categories

10.2 Colposcopic Management of CIN 1

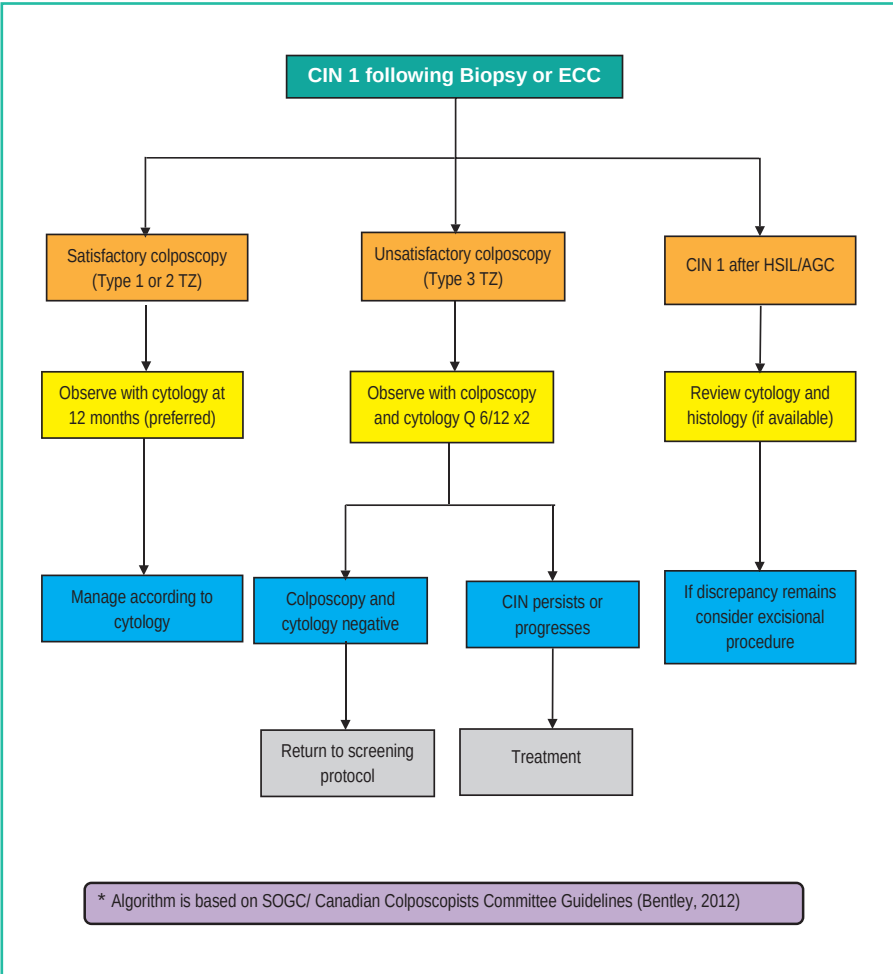


Figure 23: Management of CIN1 Following Colposcopy

10.3 Colposcopic management of CIN 2/3 in patient >25 years

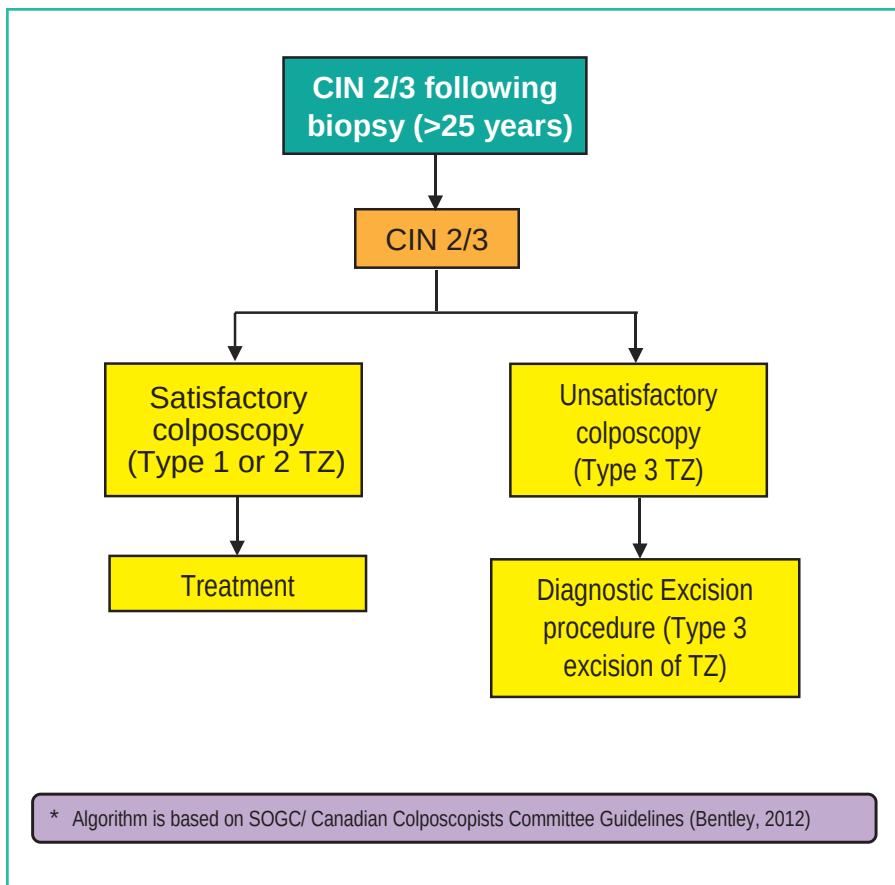


Figure 24: Management of CIN2/3 in patients aged 25 years and above following colposcopy

10.4 Colposcopic management of CIN 2/3 in patient <25 years old

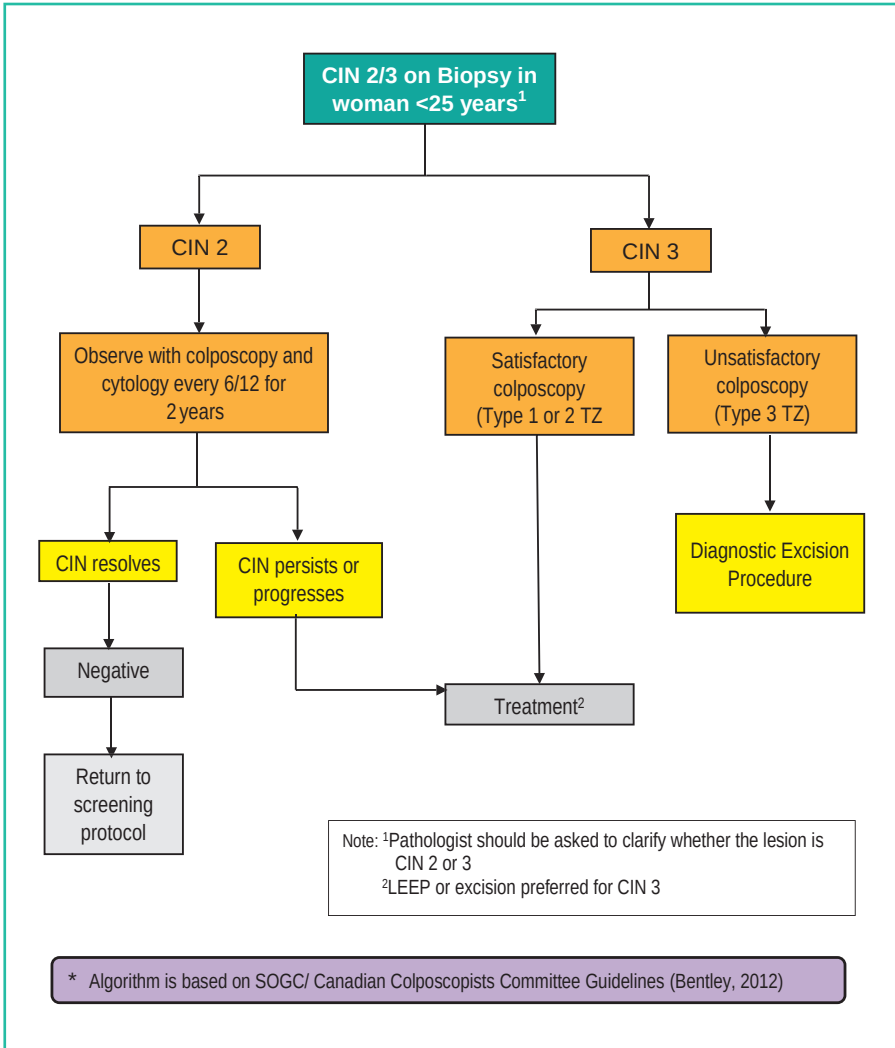


Figure 25: Management of CIN2/3 in patients aged 25 years and less following colposcopy

10.5 Follow-up post treatment for CIN

Once treated for CIN or adenocarcinoma in-situ (AIS), a woman is at risk of persistence or recurrence and at long-term risk of invasive carcinoma (Melnikow, McGahan, Sawaya, Ehlen, & Coldman, 2009; Soutter, Sasieni, & Panoskaltsis, 2006; Wright Jr et al., 2007). The aim of the follow-up is to identify any recurrence or persistence dysplasia (Bentley, 2012).

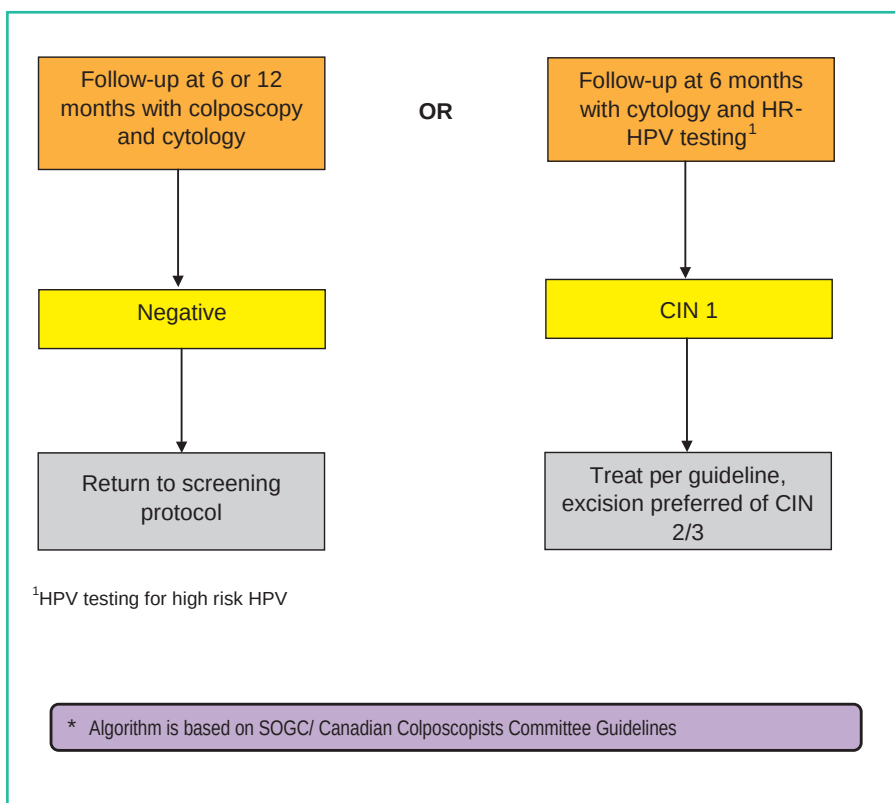


Figure 26: Options for Follow-up Post Treatment

Table below displays the recommended waiting time for colposcopy appointment:

Table 2: Recommended Waiting Time for Colposcopy Appointment Following Pap Smear Report

Type of Smear Report	Recommended Standard Waiting Time For Colposcopy Appointment	Evidence level
ASC-H or AGC	6 weeks	Level 3
HSIL/ACIS	4 weeks	Level 3
Suspicious of carcinoma	Within 2 weeks	Level 3
First time referral with other abnormal results	Within 12 weeks	Level 3



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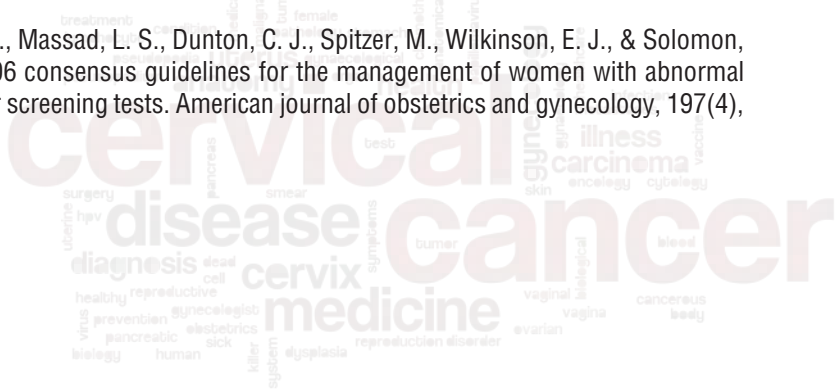
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APPENDIX 1

Definition of Terms

Terms	Definition
Colposcopy	Examination of the cervix, vagina, and, in some instances the vulva, with the colposcope after the application of a 3-5% acetic acid solution coupled with obtaining colposcopically-directed biopsies of all lesions suspected of representing neoplasia.
Endocervical sampling	Obtaining a specimen for either histopathological evaluation using an endocervical curette or a cytobrush or for cytological evaluation using a cytobrush.
Endometrial sampling	Obtaining a specimen for histopathological evaluation using an endometrial aspiration or biopsy device, a “dilatation and curettage” or hysteroscopy.
Endocervical assessment	A process of evaluating the endocervical canal for the presence of neoplasia using either a colposcope or endocervical sampling.
Diagnostic excisional procedure	A process of obtaining a specimen from the transformation zone and endocervical canal for histopathological evaluation and includes laser conization, cold-knife conization, loop electrosurgical excision procedure (LEEP), and loop electrosurgical conization.
Adequate colposcopy	Indicates that the entire squamocolumnar junction and the margin of any visible lesion can be visualized with the colposcope.

APPENDIX 2

HPV Test / Cytology Request Form

PS 1/98 (Pindaan 2019)



KEMENTERIAN KESIHATAN MALAYSIA
PERKHIDMATAN PATOLOGI

No. Makmal:

BORANG PERMOHONAN UJIAN HPV/ SITOLOGI (PAP SMEAR / LBC)
HPV TEST / CYTOLOGY (PAP SMEAR / LBC) REQUEST FORM

Hospital / Klinik Hospital / Clinic	
BUTIRAN PELANGGAN / CLIENT'S BIODATA	
i. Nama / Name	v. Alamat / Address
ii. No Kad Pengiraan / IC No	vi. No Telefon: (Rumah/ Home) Phone No (Pejabat/ Office)
iii. Etnik / Ethnicity	
iv. Umur / Age	
BUTIRAN SARINGAN / SCREENING INFORMATION (Tandakan X pada kotak berkenaan)	
i. Tarikh sampel diambil: Date sample taken	v. No. Makmal terdahulu: Previous Laboratory No
ii. Jenis Sampel: Type of sample	vi. Keputusan terdahulu: Previous diagnosis
iii. Bahagian sampel diambil: Sample site	vii. Pengambilan Sampel Oleh: Sampling by
iv. Jenis saringan: Type of screening	☐ Soritin ☐ Wangsa Kukuhan
RINGKASAN / KLINIKAL / CLINICAL SUMMARY (Tandakan X pada kotak berkenaan)	
i. Status Hormoni: Hormonal status	vi. Gejala / Tanda: Symptom / Sign
ii. Tarikh Haid terakhir: Last menstrual period	vii. Serviks: Cervix
iii. Tarikh Kelahiran terakhir: Last childbirth	viii. Maklumat tambahan: Additional information
iv. Kontracepsi (Metode hormonal): Contraceptive / hormonal therapy	- Takut Alah Sarung Plastik / HPV Damai
v. Sejarah Rawatan: Treatment history	
MAKlumat Pemohon / REQUESTING PRACTITIONER	
Nama / Name	Jawatan / CCP: Designation / Stamp
Tanda Tangan / Signature	

APPENDIX 3

Pap Smear/Liquid-based Cytology Report

KEMENTERIAN KESIHATAN MALAYSIA PERKHIDMATAN PATOLOGI FORMAT LAPORAN SITOLOGI (PAP SMEAR/LBC) PAP SMEAR/ LIQUID BASED CYTOLOGY (LBC) REPORT		05/2/2019
Name: _____		Cytology No: _____
A) Type of sample: ☐ Conventional Pap Smear ☐ Liquid-based preparation A1) Type of LBC: ☐ SurePath ☐ ThinPrep ☐ Other		
B) Sample Adequacy: ☐ i) Satisfactory for evaluation ☐ ii) Unsatisfactory for evaluation Endocervical cells / transformation zone cells ☐ Present ☐ Absent With: ☐ Obscuring blood ☐ Poor fixation / air drying artifact ☐ Thick uneven smear ☐ Thick inflammatory exudate ☐ Lack of clinical data ☐ Scanty squamous epithelial component ☐ Poor fixation / air drying artifact ☐ Obscuring blood ☐ Thick uneven smear ☐ Thick inflammatory exudate ☐ Broken slide beyond repair		
C) Interpretation / Result: ☐ i) Negative for intraepithelial lesion or malignancy (NILM) ☐ ii) Organism present: ☐ Fungal organisms, morphologically consistent with <i>Candida</i> spp. ☐ Shift in flora suggestive of bacterial vaginosis ☐ Bacteria morphologically consistent with <i>Actinomyces</i> spp. ☐ Cellular changes associated with Herpes Simplex Virus ☐ Trichomonas vaginalis ☐ Cytomegalovirus (CMV) ☐ iii) Epithelial cells abnormalities ☐ a) Squamous cell: ☐ Atypical squamous cells: ☐ of undetermined significance (ASC-US) ☐ cannot exclude HSIL (ASC-H) ☐ Low grade squamous intraepithelial lesion (LSIL) ☐ High grade squamous intraepithelial lesion (HSIL) ☐ Features suspicious for invasion ☐ Squamous cell carcinoma ☐ b) Other non-neoplastic findings: ☐ Descent cellular changes associated with: ☐ inflammation / typical repair ☐ irradiation ☐ Intrauterine contraceptive device (IUCD) ☐ Atrophy ☐ Presence of glandular cells post hysterectomy ☐ Presence of endometrial cells (in women ≥45 yrs of age) ☐ c) Glandular cells: ☐ Atypical cells (NCS) ☐ Endocervical cells ☐ Endometrial cells ☐ Glandular cells ☐ Atypical cells, favour neoplasia: ☐ Endocervical cell ☐ Glandular cells (NCS) ☐ Endocervical adenocarcinoma (in-situ) ☐ Adenocarcinoma ☐ Endocervical ☐ Endometrial ☐ Extracervical ☐ Not otherwise specified (NOS) ☐ d) Other malignant neoplasm, specify: _____		
D) Comments: _____		
E) Suggestion: ☐ Repeat LBC Pap Smear as Schedule ☐ Repeat Pap Smear 3-6 month ☐ Repeat Pap Smear after antineoplastic treatment ☐ Repeat smear after oestrogen therapy ☐ Refer FMS ☐ To get Co-surgery appointment ☐ Refer to: Gynaecology / Gynaecological Oncologist		
LAB USE ONLY		
Validated by: _____ Designation: _____ Date reporting: _____ Date Printing: _____ First Screener: _____ Second Screener: _____ Pathologist: _____ Review of previous Pap smear slide: YES / NO / IF YES State No: _____		
VALIDATION		
Result reviewed by: _____ Designation / Stamp: _____ Date Action: _____		

HPV Test Result Report

KEMENTERIAN KESIHATAN MALAYSIA PERKHIDMATAN PATOLOGI BORANG PENYEMPAHAN UJIAN ETYMOLOGI (FAP SKEAN/184)		HPV/2015
MOLECULAR VIROLOGY TEST RESULT REPORT		
Human Papillomavirus (HPV) Genotyping		
NAME		
NRIC NO.		
REGISTRATION NO.		
WARD / KLINIK		
HOSPITAL		
NEGERI		
REQUESTED BY		
SAMPLE DETAILS FOR MOLECULAR VIROLOGY TEST		
Sample Type	☐ Self sample ☐ SW	Date of Test
Sample Collection Method		Lab Barcode No.
Sample Transport Condition	☐ Ice ☐ Dry	Date of Sample Collection
		Date of Sample Received
RESULT DETAILS FOR MOLECULAR VIROLOGY TEST		
QUANTITATIVE Real-time PCR - HPV VIRUS DNA		
Overall Interpretation: (Target X probe both detected)	HPV	Detected
	TR	Not Detected
	HPV	Detected
	TR	Not Detected
	HPV	Detected
	TR/HPV	Not Detected
		INVALID / UNSATISFACTORY
NOTES :		
a) Test method : Real-Time Polymerase chain reaction (RT-PCR).		
b) Genotype detection : TR, TR and other High Risk HPV DNA (71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89).		
c) "Not Detected" result does not conclusively rule out the agent tested, for the following reasons:		
+ Specimens collection not timely.		
+ Specimen deterioration / breakdown of viral shells during storage or transportation.		
+ Detection value lower than limit of detection.		
d) This result shall be interpreted in conjunction with other clinical laboratory findings.		
e) This report shall not be reproduced except with written approval from the laboratory.		
COMMENTS :		
SUGGESTION		
	Repeat HPV test at 6 years (HPV not detected).	
	Repeat HPV test within 12 months (Insatisfactory HPV test).	
	For Liquid Based cytology (LBC) (HPV Positive case 18/18).	
	Refer for COLPOSCOPY (HPV Positive 18/18).	
Authorized by:		
Date:		

APPENDIX 5A

Notification Letter to Clients with Normal Smear / HPV Negative (English)

Name :

Address :
.....

Date :

Mrs/Ms,

Your HPV / Pap smear / LBC has been reported as normal. You are advised to undergo a repeat HPV / Pap smear / LBC on

However, if you should experience any symptoms such as vaginal bleeding after intercourse or vaginal discharge, please consult your doctor.

Signature

Name :

Department chop :

** Kindly take note that a HPV / Pap smear / LBC test is a screening test. In a small number of women this test can fail to detect abnormal cervical changes.*



APPENDIX 5B

**Notification Letter to Clients with Normal Smear / HPV Negative
(Malay)**

Nama :

Alamat :
.....

Tarikh :

Puan,

Ujian HPV / Pap smear / LBC anda telah dilaporkan sebagai normal. Anda
dinasihatkan untuk melakukan ujian HPV / Pap smear / LBC pada
.....

Walaupun bagaimanapun, sekiranya puan mengalami sebarang tanda seperti discaj
vagina atau pendarahan vagina selepas persetubuhan, sila berjumpa doktor.

.....
Tandatangan

Nama :

Cop Jabatan :

** Sila ambil perhatian bahawa ujian HPV / Pap smear / LBC adalah satu ujian
saringan. Dalam sebilangan kecil wanita ujian ini gagal mengesan tanda-tanda
abnormal pada/dalam pangkal rahim.*

APPENDIX 5C

Notification Letter to Clients with Abnormal HPV / Pap Smear / LBC (English)

Name :

Address :
.....

Date :

Mrs/Ms,

Your HPV / Pap smear / LBC test result is ready. You are kindly requested to come to the clinic for follow-up treatment as follows:

Date :
Time :
Clinic / Hospital :

Thank you.

.....
Signature

Name :
Department chop :



APPENDIX 5D

Notification Letter to Clients with Abnormal HPV / Pap Smear / LBC (Malay)

Nama :

Alamat :
.....

Tarikh :

Puan,

Keputusan ujian HPV / Pap smear / LBC puan telah siap. Puan dikehendaki datang ke klinik untuk mendapatkan rawatan selanjutnya sebagaimana berikut:

Tarikh :
Masa :
Klinik/ Hospital :

Terima kasih.

.....
Tandatangan

Nama :
Cop Jabatan :

APPENDIX 6

Cervical Brush for Cervical Sampling



APPENDIX 7

A guide and reporting proforma for colposcopic biopsies performed for the diagnosis and treatment of pre-invasive cervical neoplasia.

[For invasive carcinoma please refer to the Clinical Practice Guidelines on management of Cervical Cancer (Second Edition) MOH/P/PAK/294.15(GU)]

1. CLINICAL INFORMATION		
1.1 Procedure performed	• Cervical biopsy a. Cervical excision type -Type 1 (at least 6mm up to 10mm length) -Type 2 (10 to 15mm length) -Type 3 (>15mm length) b. Cervical excision modality -Electrosurgical excision eg Loop excision such as LLETZ, LEEP -Cold-knife cone biopsy -Laser cone biopsy • Endocervical curettage • Other, specify	Note: both type of excision and modality should be provided (for audit purposes).
1.2 Colposcopic findings		
1.3 Location of any lesions		
1.4 HPV results	• HPV 16 detected • HPV 18 detected • Oncogenic HPV (not 16/18) detected • Oncogenic HPV not detected • Other, specify	Other relevant information eg: date of testing
1.5 LBC results		
1.6 Relevant gynaecological procedure or treatment		
1.7 Comments		

2. MACROSCOPIC		
2.1 Specimen labelled as		
2.2 Type of procedure	• Cervical biopsy a. Cervical excision type -Type 1 (at least 6mm up to 10mm length) -Type 2 (10 to 15mm length) -Type 3 (>15mm length) b. Cervical excision modality -Electrosurgical excision eg Loop excision such as LLETZ, LEEP -Cold-knife cone biopsy -Laser cone biopsy • Endocervical curettage • Other, specify	Note: both type of excision and modality should be provided (for audit purposes).
2.3 Orientation markers	Not present / Present, specify	For excision specimens
2.4 Specimen measurement		
	Number of pieces – PLEASE INDICATE	Report if more than 1 piece is received.
	Maximum dimension Numeric: ____mm	Report for Small Diagnostic Biopsies <i>Note:</i> repeat for each piece received
	Dimensions Numeric: ____x____x____mm	Report for non-conical excision specimens and the below measurements are not possible. <i>Note:</i> repeat for each piece received
	Length of canal Numeric: ____mm	Report for conical excision specimens <i>Note:</i> This is measured from the external os to the apex.

	Length of specimen Numeric: ____mm	Report for conical excision specimens
	Diameter of ectocervix Numeric: ____x ____mm	Report for disorientated, conical Loop/Laser excisions
	Diameter of ectocervix in 3-9 o'clock plane Numeric: ____mm	Report for orientated, conical excision specimens
	Diameter of ectocervix in 6- 12 o'clock plane Numeric: ____mm	Report for orientated, conical excision specimens
	Macroscopically visible lesions Present or Absent Other macroscopic comment	For excision specimens
	Sampling (blocks).	Eg: Serially sliced and submitted in entirety as in the diagram

3. MICROSCOPIC FINDINGS

3.1 Tissues present	• Endocervical mucosa • Squamous mucosa • Other, specify	
3.2 Tissue artefact	Absent Thermal • Minimal • Moderate • Extensive, impacting margin assessment • Extensive, impacting diagnostic assessment • Extensive, impacting both margin and diagnostic assessment. Non-thermal artefact present, specify	• minimal – thin rim of thermal artefact only with no significant interference with histological assessment • moderate – thermal artefact focal or partially hinders histological assessment • extensive – thermal artefact significantly interferes with histological

		assessment Specify whether thermal artefact interferes with: - o margin assessment: if known the specific margin(s) affected should be documented. Whilst caution is advised in interpretation in tissue affected by artefact, when there is significant thermal artefact at a tissue margin, p16 IHC may assist in clarifying margin assessment. - o diagnostic assessment – for example grading of SIL, inability to exclude invasion, diagnostic certainty of AIS.
3.3 Degree of epithelial loss	• Minimal • Moderate • Extensive, impacting margin assessment • Extensive, impacting diagnostic assessment • Extensive, impacting both margin and diagnostic assessment	• minimal – there is no significant epithelial loss (<10%) • moderate – focal or partial epithelial loss that may hinder histological assessment (10-30%) • extensive – significant epithelial loss that is likely to hinder histological

		assessment (>30%) Specify as to whether epithelial loss influences: - o margin assessment: if known, specific margin(s) affected should be documented - o diagnostic assessment – if there is a specific known diagnostic issue this should be documented, however generally significant epithelial loss results in the possibility of a missed diagnosis that is unable to be further qualified.
3.4 CIN LESION	Present Possible Not identified	
3.5 CIN LESION subtype	CIN1 CIN2 CIN3	CIN1 (Applicable only if no HSIL/AIS or invasive carcinoma is present)
3.6 CIN LESION EXTENT		For excision specimens
3.7 Endocervical Adenocarcinoma in situ (AIS)	Present Possible Not identified	
3.8 AIS Extent		For excision specimens

APPENDIX 8

Examples of Pathology Report

Example Report 1

CLINICAL INFORMATION PROVIDED Procedure performed: Colposcopic findings: LBC results: Relevant gynaecological procedure or treatment: MACROSCOPIC Specimen labelled as: Maximum dimension: Specimen description: MICROSCOPIC Tissues present: Tissue artefact: Degree of epithelial loss: CIN2/3: CIN1: Endocervical Adenocarcinoma in situ (AIS): Other microscopic comment: ANCILLARY TESTS: CASE CATEGORISATION Squamous component: Endocervical component:	Cervical biopsy LSIL LSIL LLETZ in 2012 Cervical biopsy 12 o'clock 5mm Pale tissue Squamous mucosa; Endocervical mucosa Absent Minimal Not identified Present Not identified Features of HPV effect Not performed Low-grade Squamous Intraepithelial Lesion (LSIL) Normal
DIAGNOSTIC SUMMARY Cervical biopsy – Low Grade Squamous Intraepithelial Lesion (LSIL)	

Example Report 2

CLINICAL INFORMATION PROVIDED Procedure performed: Colposcopic findings: HPV results: LBC results: MACROSCOPIC Specimen labelled as: Orientation markers: Number of pieces: SPECIMEN SIZE Piece 1 dimensions: Piece 2 dimensions: Specimen description: Macroscopically visible lesions: Sampling: MICROSCOPIC Tissues present: Tissue artefact: Degree of epithelial loss: CIN2/3: CIN1: Endocervical Adenocarcinoma in situ (AIS): ANCILLARY TESTS: Immunohistochemistry:	LLETZ Cervical Normal HPV 16/18 detected ASC-H LLETZ Not present 2 12x9x4mm 11x8x4mm Tan rubbery tissue with a small amount of mucosa Absent 1.1–1.3 larger piece serially sectioned with 1.1 containing end pieces, 1.4-1.6 smaller piece serially sectioned with 1.4 containing end pieces Squamous mucosa; Endocervical mucosa Thermal, extensive, impacting diagnostic assessment Extensive, impacting diagnostic assessment Not identified Not identified Not identified Performed P16 immunohistochemistry negative on blocks 1.2 and 1.5
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CASE CATEGORISATION Squamous component: Endocervical component: Comment:	Benign findings Benign findings Severe thermal artefact hinders accurate histological assessment
DIAGNOSTIC SUMMARY Cervical biopsy – Low Grade Squamous Intraepithelial Lesion (LSIL)	

Example Report 3

CLINICAL INFORMATION PROVIDED Procedure performed: Colposcopic findings: Location of any lesions: HPV results: LBC results: Relevant gynaecological procedure or treatment:	Cervical punch biopsy ?HSIL ?metaplasia 6 o'clock HPV 18 detected ASC-H Colposcopy / biopsy
MACROSCOPIC Specimen labelled as: Maximum dimension: Specimen description:	Cervical biopsy 6 o'clock 4mm Pale tissue Endocervical mucosa
MICROSCOPIC Tissues present: CIN2/3: CIN1: Endocervical Adenocarcinoma in situ (AIS): Other relevant pathology: Other microscopic comment:	Endocervical mucosa Not identified Not identified Nil - The biopsy was examined at multiple (12) levels, all showing benign endocervical glandular mucosa only. No squamous epithelium was seen in the multiple levels.

ANCILLARY TESTS: CASE CATEGORISATION Squamous component: Endocervical component: Comment:	There are no features of cervical glandular neoplasia. Not performed Not identified Normal The biopsy was benign, however no squamous mucosa was identified.
DIAGNOSTIC SUMMARY Cervical biopsy 6 o'clock – Benign endocervical mucosa	

Example Report 4

CLINICAL INFORMATION PROVIDED Procedure performed: Colposcopic findings: Location of any lesions: HPV results: LBC results: Relevant gynaecological procedure or treatment: MACROSCOPIC Specimen labelled as: Maximum dimension: Specimen description: MICROSCOPIC Tissues present: Degree of epithelial loss: CIN2/3: Endocervical Adenocarcinoma in situ (AIS): ANCILLARY TESTS:	Cervical biopsy Favour LSIL 12 o'clock HPV 16 detected HSIL - Cervical biopsy 12 o'clock 5mm Pale grey tissue Block A: submitted entirely Squamous mucosa; Endocervical mucosa Minimal Present CIN2 Not identified.
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CASE CATEGORISATION Squamous component: Endocervical component: Comment:	Performed p16 positive (strong, block-like) Cervical Intraepithelial Neoplasia 2 (CIN 2) Normal On H&E sections there are prominent features of HPV effect, and atypical squamous epithelial cells extending mid-way through the squamous mucosa. As p16 immunohistochemistry is positive, this is classified as CIN 2.
DIAGNOSTIC SUMMARY Cervical biopsy – Cervical Intraepithelial Neoplasia 2 (CIN 2)	



FAMILY HEALTH DEVELOPMENT DIVISION
MINISTRY OF HEALTH MALAYSIA